

Metabolic Basis of Solute Transport

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Metabolism refers to the entire set of interconnected chemical reactions within living organisms that form and maintain tissue and govern the storage and release of energy in order to sustain life. This chapter is dedicated to one aspect of kidney metabolism—namely, the storage, release, and utilization of energy by the nephron as it transforms the glomerular filtrate into urine.

How much energy is required to make the urine? Of the major body organs, the kidney consumes the second highest amount of oxygen per gram of tissue (2.7 mmol/kg/min vs. 4.3 mmol/kg/min for the heart).¹ Most of the potential energy provided by renal oxidative metabolism is committed to epithelial transport, which determines the volume and composition of the urine. It has been asserted that, because the kidney reabsorbs 99% of the glomerular filtrate, it must use a lot of energy. But this logic is incorrect. The minimum net energy required for reabsorption does not depend on the amount of fluid that is reabsorbed. Forming a volume of urine with a solute composition equal to that of the body fluid from which it is formed is the thermodynamic equivalent of partitioning a bucket into two compartments by the use of a divider, which requires no net energy. On the other hand, energy is required to form a urine that differs in solute composition from that of the body fluids (i.e., plasma). To appreciate this, consider that the hypothetical remixing of urine with plasma would cause the formation of entropy, known as mixing entropy. Thus, energy is required to form urine from plasma and attain a state of reduced entropy. The minimum amount of energy required for this is equal to the temperature multiplied by the decrease in mixing entropy associated with the differential solute composition of urine versus plasma.

This chapter provides an overview of the interdependence of renal solute transport and renal metabolism, including (1) the role of the sodium pump, $\text{Na}^+\text{-K}^+\text{-adenosine triphosphatase}$ (ATPase), in epithelial transport; (2) the metabolic substrates

fueling active transport along the nephron and regional metabolic considerations; (3) the role of renal blood flow, the glomerular filtration rate (GFR), and tubuloglomerular feedback in controlling fluid and electrolyte filtration and tissue oxygenation; (4) the amount of oxygen consumed per sodium reabsorbed ($\text{QO}_2/\text{T}_{\text{Na}}$); and (5) the metabolic efficiency of transport during normal perturbations and disease.

THERMODYNAMIC APPROACH TO METABOLISM AND TRANSPORT

THERMODYNAMIC ANALYSIS OF KIDNEY FUNCTION

Interest in kidney metabolism antedates most knowledge of the kidney's inner workings or of biochemistry. The theoretical minimum amount of energy required to make urine was determined from the laws of equilibrium thermodynamics nearly a century ago. For a human in balance on a typical diet, the cost of converting the glomerular filtrate into urine by an idealized process that is 100% efficient, infinitely slow, completely reversible, involves no back-leak, and generates no entropy and heat is about 0.5 cal/min/1.73 m².² In reality, the kidney consumes more than 50-fold this amount of energy. On this basis alone, one might argue that the kidney is horribly inefficient, even after one subtracts the cost of the kidney maintaining itself. On the other hand, added costs are imposed by the requirement to make urine in a finite amount of time, the need for flexibility to rapidly alter the volume and composition of the urine, the stoichiometric constraints of biochemistry, the known limits on the thermodynamic efficiency of oxidative phosphorylation, and the intrinsic permeabilities of tissues to electrolytes, gases, and urea.

The thermodynamic requirement may be a small fraction of the actual expenditure, but before one concludes that

the body is unconcerned with thermodynamics, it may be noted that the thermodynamic energy required of the kidney to maintain salt and nitrogen balance with consumption of a typical diet is minimized with the usual water intake of 1 to 2 L/day. This suggests that the human body evolved to minimize the thermodynamic energy requirements of the kidney.

Moreover, the thermodynamic cost of excreting urea declines as blood urea nitrogen (BUN) concentration increases. Thus, as BUN rises in kidney disease, less energy is required to maintain the nitrogen balance. In kidney disease, the urine composition is also restricted to a narrower range. Using a classical thermodynamic approach, Newburgh suggested that the composition of the urine and the body fluids in kidney disease is determined by the available free energy; he noted that the declining flexibility of the diseased kidney to vary the urine composition could be predicted from the reduced free energy available for transport.³

APPLICATION OF THE LAWS OF THERMODYNAMICS TO KIDNEY FUNCTION

The macroscopic laws of equilibrium thermodynamics apply to kidney metabolism; any theory of metabolism is necessarily incorrect if it violates these laws. The laws of thermodynamics essentially describe transitions of a system from one state to another. The first law of thermodynamics states that total energy is conserved during any process that occurs in a closed system. When a system is open to its environment, the combined energy of the system + environment remains constant. When the total internal energy, temperature, pressure, and volume of a system remain constant, any process that yields a change in free energy also yields reciprocal changes in entropy. Doing work on the system is equivalent to adding free energy to the system, which determines the upper limit of how much useful work the system can do against its environment.

The first law stipulates that total energy is conserved throughout any process but provides no other indication of whether a given process will occur spontaneously. The glomerular filtrate contains a mixture of salt and urea. The tubule partitions this into urine and reabsorbate. The urine has a different ratio of urea to salt than the reabsorbate, so the entropy has decreased. But the total internal energy of the combined urine and reabsorbate is the same as the original filtrate. Hence, the first law would be satisfied if the urine were to form spontaneously from the filtrate. The fact that NaCl and urea never sort themselves spontaneously into regions of higher and lower concentration is a consequence of the second law of thermodynamics, which states that all spontaneous processes generate entropy. Conversely, all spontaneous processes dissipate free energy and will cease when the supply of free energy is exhausted. It is possible to reduce entropy or elevate free energy in a system, but only if the system imports energy from its surroundings, in which case, there will be an increase in entropy of the surroundings that exceeds the decrease in entropy of the system. Some processes in the kidney, such as conversion of chemical to mechanical energy by the Na⁺-K⁺-ATPase, are highly efficient and generate almost no entropy. Other processes, such as the countercurrent multiplier, are inefficient and generate a lot of entropy. As a rule, those processes

that generate the least entropy work over short distances and short times.

The laws of equilibrium thermodynamics determine the direction of any spontaneous process, but they do not address the rate of change. Hence the laws of equilibrium thermodynamics are not adequate for a full description of a living system that is displaced from equilibrium and characterized by flow of matter and energy within the system itself, as well as between the system and its environment. Thermodynamic principles are extended to incorporate time as a variable by the theory of nonequilibrium thermodynamics. Nonequilibrium thermodynamics entails certain assumptions and approximations that make it more of a tool and less of an edifice than classical equilibrium thermodynamics, but the theory performs well in many areas of physiology, including transport physiology. Basically, the theory asserts that the flow of any extensive property (e.g., mass, volume, charge) is the product of a driving force and a proportionality constant, which has units of conductance. It applies to both macro- and micro-processes involved in forming the urine. Examples include all mechanisms for secondary active transport and the conversion of chemical to translational free energy by ATPases.

ENERGY AND THE SODIUM PUMP

Na⁺-K⁺-ATPase, also referred to as the sodium pump, is a ubiquitous plasma membrane protein that transports intracellular sodium out of the cell and extracellular potassium into the cell, thereby generating opposite concentration gradients for sodium and potassium ions across the cell membrane. This process of separating sodium from potassium across the cell membrane is fueled by the hydrolysis of adenosine triphosphate (ATP).^{4,5} Each cycle of the pump consumes 1 ATP molecule while transporting 3 Na⁺ and 2 K⁺ ions across the cell membrane. The hydrolysis of ATP and the associated transport of ions are mutually dependent^{4,5} and constitute an example of primary active transport. In this process there is nearly full conversion from chemical to mechanical energy, with minimal dissipation. The translational energy that develops after ATP hydrolysis results from electrostatic repulsion between the product ions, ADP and Pi, in accordance with Coulomb's law. Although this energy could be dissipated through subsequent collisions, such events are unlikely over very short time scales and short distances. For a relative kinetic energy of the phosphate of 0.6 eV, for example, the phosphate ion moves about 0.1 nm in 0.3 ps. If no other collisions occur in that short time interval, the phosphate can then transfer its entire kinetic energy to the sodium pump in the form of a molecular strain. Given the intrinsic free energy of ATP hydrolysis, the pump can generate gradients that store up to approximately 0.6 eV of electrochemical potential per 3 Na⁺ plus 2 K⁺ ions. For a typical cell in a typical environment, about 0.4 eV is required to cycle the pump against the existing Na and K gradients, which means that cells tend to operate with some reserve to further reduce their sodium or increase their potassium concentrations.

STRUCTURE OF THE SODIUM PUMP

The sodium pump is composed of an α catalytic subunit, which hydrolyzes ATP and transports Na⁺ and K⁺ across the

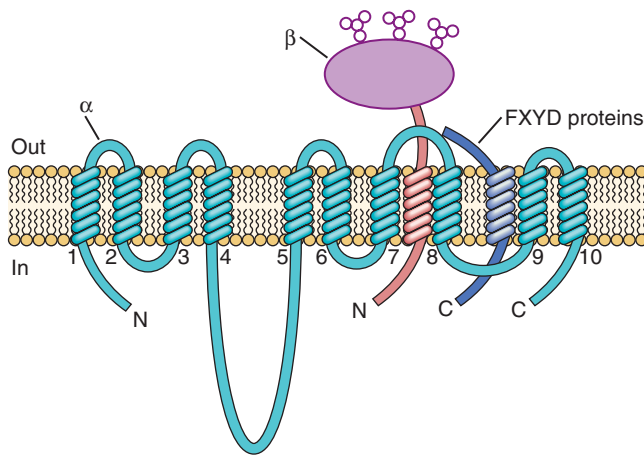


Fig. 5.1 Na⁺-K⁺-ATPase is composed of a catalytic α -subunit (teal), an obligatory β -subunit (pink), and tissue-specific FXYD proteins (blue). The α -subunit has 10 transmembrane segments. It hydrolyzes ATP, is phosphorylated in the large cytoplasmic loop, and transports sodium and potassium. The β -subunit is a type II glycoprotein that is located close to M7/M10 and interacts with the extracellular loop between transmembrane segments M7 and M8 and with intracellular regions of the α -subunit.³ FXYD proteins are type I membrane proteins that interact with M9 with the β -subunit,⁴ and in the case of FXYD1 with the intracellular lipid surface and the cytoplasmic domain of the α -subunit [5•]. (From Geering K. Functional roles of Na,K-ATPase subunits. *Curr Opin Nephrol Hypertens*. 2008;17[5]:526–532.)

membrane, a β -subunit that is critical for functional maturation and delivery of Na⁺-K⁺-ATPase to the plasma membrane, and an FXYD protein that can modulate the kinetics of Na⁺-K⁺-ATPase in a tissue-specific manner⁶ (Fig. 5.1). There are multiple isoforms of each subunit. The $\alpha 1\beta 1$ heterodimer is likely the exclusive Na⁺-K⁺-ATPase in renal epithelia,⁷ whereas several FXYD protein subunits are expressed differentially along the nephron.^{6–9} Biophysical models describing the turnover of the sodium pump through its functional cycle are described in a review by Horisberger.⁴

OTHER ADENOSINE TRIPHOSPHATASES

Besides Na⁺-K⁺-ATPase, additional ion-translocating ATPases are expressed in renal epithelia along the nephron,¹⁰ including H⁺-K⁺-ATPase,^{11,12} Ca²⁺-ATPases,¹³ and H⁺-ATPases.^{14,15} These transport ATPases play important roles in maintaining urinary acidification and calcium homeostasis as discussed in Chapters 6, 7, and 9. These ATPases do not contribute significantly to the reabsorption of the bulk of the filtrate.

PUMP LEAK PROCESS AND THE SODIUM POTENTIAL

For a cell in a steady state, the pumping of ions by the Na⁺-K⁺-ATPase must be offset by an equal and opposite diffusion of those ions back across the cell membrane. The back-leak of ions is an example of electrodiffusion. This diffusion of ions generates an electric field to retard diffusion of the most mobile charged species, thereby transferring free energy from the chemical potential of the mobile species to an electrical potential acting on the less mobile species.

If the electric field is constant within the cell membrane, then the electrical potential difference across the membrane is given by the Goldman voltage equation, which is shown here for a membrane that is permeable to Na, K, and Cl:

$$\psi = \frac{RT}{F} \ln \left(\frac{P_K[K]_o + P_{Na}[Na]_o + P_{Cl}[Cl]_o}{P_K[K]_i + P_{Na}[Na]_i + P_{Cl}[Cl]_i} \right)$$

where P_X is the permeability to X, $[X]_o$ is the concentration of X outside the cell, and $[X]_i$ is the concentration inside the cell. If the permeability to one ion dominates the others, then the membrane voltage approaches the Nernst potential for that ion and the free energy is transferred to electrochemical potential of the other ions. If chloride is not actively transported, then the second law of thermodynamics dictates that no free energy exists in the chloride gradient. Thus, for a membrane that actively transports Na and K and is primarily permeable to K, the membrane voltage approaches the Nernst potential for K and the free energy provided by active transport is all transferred to the transmembrane Na difference.

To summarize, because cell membranes are generally more permeable to potassium than to sodium, potassium diffusion contributes more to the cell voltage than sodium diffusion, even though three sodium ions leak into the cell for every two potassium ions that leak out. Thus, diffusion of potassium out of the cell dominates the cell voltage, making it negative. The negative cell voltage, in turn, neutralizes the net driving force for further potassium egress and augments the net driving force for sodium entry. Because cell membranes are poor capacitors, an imperceptible charge imbalance suffices to form the entire membrane voltage. This allows the transmembrane concentration differences for sodium and potassium to remain nearly equal and opposite despite the much greater permeability to potassium. The net outcome of this pump-leak process is that electrochemical potential, which originates with ATP hydrolysis, becomes concentrated in the transmembrane sodium gradient, whereas potassium resides near electrochemical equilibrium.

HARNESSING THE SODIUM POTENTIAL FOR WORK

The difference in electrochemical potential for sodium across the cell membrane is available to drive the unfavorable passage of other solutes across the membrane by a variety of exchangers and cotransporters. Examples include the proximal tubule Na⁺/H⁺ exchanger, sodium-glucose cotransporters (SGLTs), the basolateral Na/ α -ketoglutarate (α -KG) cotransporter, the furosemide-sensitive Na-K-2Cl cotransporter, (NKCC2), and the thiazide-sensitive Na-Cl cotransporter (NCC). Generically, transport that directly uses free energy from the sodium gradient to drive uphill flux of another solute is referred to as secondary active transport¹⁶ (α -KG cotransport in Fig. 5.2). Tertiary active transport refers to the net flux of a solute against its electrochemical potential gradient coupled indirectly to the Na⁺ gradient (three transport processes working in parallel). An example of tertiary active transport is the uptake of various organic anions from the peritubular blood into the proximal tubular cell by the so-called organic anion transporters (OATs). Energy from the sodium gradient is converted into a gradient for α -KG to diffuse out of the cell

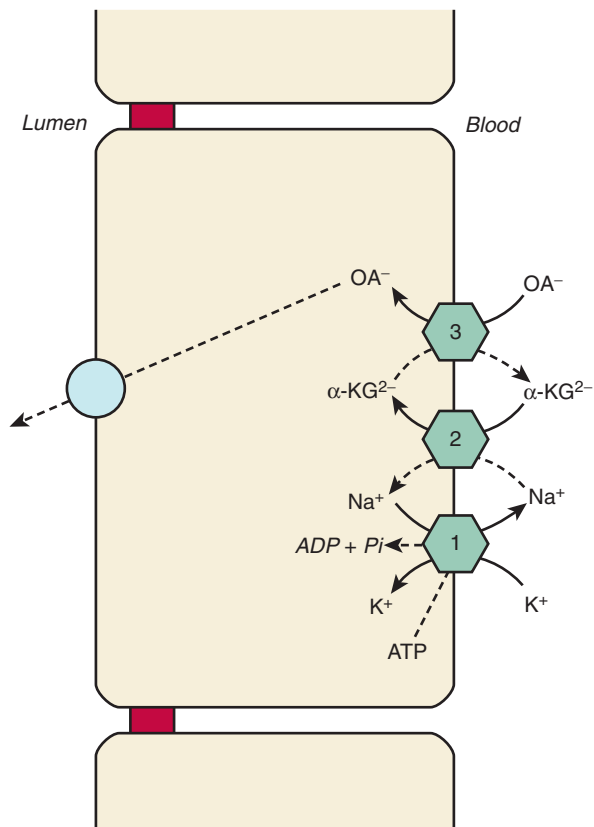


Fig. 5.2 Different modes of active uphill transport as exemplified by organic acid (OA) secretion in proximal tubule epithelial cells. Transport across the basolateral membrane involves three steps functioning in parallel. Primary active transport (1) of Na and K by $\text{Na}^+\text{-K}^+\text{-ATPase}$ coupled to the hydrolysis of ATP establishes the inwardly directed Na gradient. Secondary active transport (2) of α -ketoglutarate (α -KG) with Na on an Na/ α -KG cotransporter uses the inwardly directed Na gradient to drive α -KG into the cell. Tertiary active transport (3) of OA with α -KG on an OA/ α -KG antiporter uses the outward downhill transport of α -KG to drive the inward uphill transport of OA. The α -KG is recycled through the Na/ α -KG cotransporter, which thus links the uphill transport of OA to the generation of the Na gradient by the $\text{Na}^+\text{-K}^+\text{-ATPase}$. Ultimately OAs are secreted down the OA concentration gradient into the tubular lumen. (From Dantzler WH, Wright SH. The molecular and cellular physiology of basolateral organic anion transport in mammalian renal tubules. *Biochim Biophys Acta*. 2003;1618[2]:185–193.)

by Na/ α -KG cotransport. OATs use this potential difference to exchange α -KG for another organic anion¹⁷ (see Fig. 5.2).

For tubular cells that actively reabsorb chloride, free energy is transferred from the Na potential to drive apical chloride entry and raise cell chloride above equilibrium. In the proximal tubule, the energy for apical chloride entry is derived circuitously via sodium-hydrogen exchange that is coupled to oxalate, formate, or hydroxyl ion transport (see Chapter 5). In the thick ascending loop of Henle (TALH) and distal convoluted tubule (DCT), the energy transfer occurs by direct cotransport with Na via NKCC2 or NCC. In each case, raising cell chloride above equilibrium provides a driving force for chloride to diffuse out of the cell across the basolateral membrane, which is permeable to chloride. Raising cell chloride also makes the basolateral membrane voltage less negative, as is apparent from the Goldman equation. Because luminal voltage is the sum of voltage steps across the

basolateral and apical membranes, raising cell chloride in a cell with basolateral chloride conductance will raise the lumen voltage (make it more positive), thus providing free energy that can either be dissipated by the intercellular back-leak of chloride, which would increase entropy, or be applied to do the useful work of cation reabsorption, which would decrease entropy. The kidney uses the latter mechanism of energy transfer to augment Na reabsorption in the proximal tubule as well as calcium and magnesium reabsorption in the TALH.

For cells that express ENaC, opening these channels will depolarize the apical membrane, as can be seen from the Goldman equation. K ions, which enter the cell via the basolateral Na pump, can leave the cell by K conductances in either basolateral or apical membranes. Depolarizing the apical membrane will increase the fraction of K ions leaving by way of the apical membrane conductance. This represents the transfer of free energy from the $\text{Na}^+\text{-K}^+\text{-ATPase}$ and the apical Na potential to the useful work of K secretion.

CELL POLARITY AND VECTORIAL TRANSPORT

The polar arrangement of transporters in renal cells is essential for vectorial transport. Wherever it is expressed along the nephron, the sodium pump, which removes sodium from the cell, is restricted to the basolateral membrane. Meanwhile, the variety of exchangers, cotransporters, and sodium channels through which sodium enters the tubular cell are restricted to the apical membrane. These include the principal Na^+/H^+ exchanger (NHE3) and SGLTs in the proximal tubule, the NKCC2 in the thick ascending limb (TAL) of the loop of Henle, the NCC in the distal convoluted tubule, and epithelial sodium channels in the connecting tubule and collecting duct (see Chapter 5). These apical sodium transporters effect secondary active transport coupled to the primary active transporter, $\text{Na}^+\text{-K}^+\text{-ATPase}$.

Close coordination of sodium uptake across the apical membrane with sodium extrusion across the basolateral membrane is required to avoid osmotic swelling and shrinking of the cell. Assuming ATP is not limiting for basolateral exit, the magnitude of transepithelial transport is a function of (1) the number of transporters in the plasma membrane, which can be varied by changes in synthesis or degradation rates and/or trafficking between intracellular and plasma membranes, and (2) the activity per transporter, which can be varied by covalent modification (e.g., phosphorylation or proteolysis) or protein-protein interaction (e.g., $\text{Na}^+\text{-K}^+\text{-ATPase}$ kinetics are influenced by FXYD subunit association).⁶ The rate of apical sodium entry is also subject to influence by the availability of substrates for cotransport. For example, the amount of sodium-glucose cotransport depends on the availability of glucose in proximal tubular fluid, and the sodium entry at a given point along the TAL is subject to variations in the local chloride concentration, because NKCC2 has a relatively low affinity for chloride.

Many factors and hormones known to regulate renal sodium reabsorption (including angiotensin II, aldosterone, dopamine, parathyroid hormone, and blood pressure) act in parallel to affect the activity, distribution, or abundance of apical transporters and basolateral sodium pumps.^{7,18} The

molecular basis of this apical–basolateral crosstalk is not clearly understood, especially in the light of close cell volume control; however, there is evidence for a role of elevated cellular calcium level in response to depressed sodium transport.¹⁹ There is also recent evidence for a salt-inducible kinase that responds to slight elevations in cell Na and Ca,²⁰ as well as evidence for coupling of Na⁺-K⁺-ATPase to apical channel activity.²¹

METABOLIC SUBSTRATES FUELING ACTIVE TRANSPORT ALONG THE NEPHRON

Mitchell has noted that:

“Biochemists generally accept the idea that metabolism is the cause of membrane transport.”

The underlying idea of the hypothesis put forward here is that if the processes that we call metabolism and transport represent events in a sequence, not only can metabolism be the cause of transport, but also transport can be the cause of metabolism. Thus, we might be inclined to recognize that transport and metabolism, as usually understood by biochemists, may be conceived advantageously as different aspects of one and the same process of vectorial metabolism.²²

METABOLISM BASICS

Detailed accounts of cellular metabolism are provided in many excellent texts²³; nonetheless, an abbreviated overview

relevant to renal metabolism is warranted. Substrates enter the kidney by renal blood flow (RBF) and GFR and enter renal epithelial cells by substrate transporters, often facilitated by the inward-directed Na⁺ gradient created by the sodium pump (see Fig. 5.2), as discussed thoroughly in Chapter 8. Oxygen is likewise delivered by RBF to the epithelial cells. Once in the cell, substrates face one of three fates: (1) transport across the epithelium back into the blood (reabsorption); (2) conversion into another substrate (e.g., lactate to pyruvate); or (3) oxidation to CO₂ in the process of cellular ATP production.²⁴ This section traces the roadmap that connects substrates to production of ATP in the mitochondrion and to ATP utilization by the sodium pump, and the feedback connections between production and utilization.

Renal epithelia, except in the descending and thin ascending limbs of the loop of Henle, are packed with mitochondria (see Chapter 2). All the pathways of fuel oxidation take place in the mitochondrial matrix, except for glycolysis, which occurs in the cytosol. Substrates in the cytosol can freely cross the outer mitochondrial membrane through integral membrane porins. These substrates, as well as adenosine diphosphate (ADP) and phosphate (the building blocks of ATP), cross the inner mitochondrial membrane into the mitochondrial matrix via specific substrate transporters driven by their respective concentration gradients or by the H⁺ gradient created by the electron transport chain (ETC; Fig. 5.3).

As illustrated in Fig. 5.4, amino acids, fatty acids, and pyruvate are metabolized to acetyl-coenzyme A and enter the citric acid cycle. With each turn of the cycle, three molecules of reduced nicotinamide adenine dinucleotide

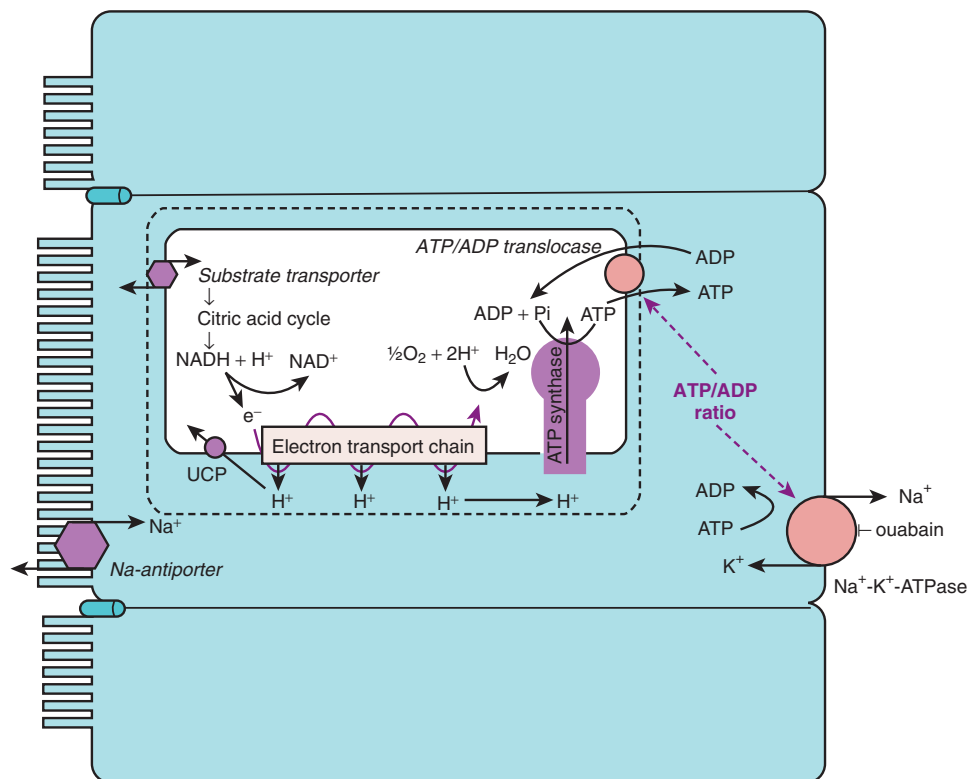


Fig. 5.3 Whittam model. Coupling of ATP utilization by Na⁺-K⁺-ATPase to ATP production by mitochondrial oxygen consumption (QO₂). Hydrolysis of ATP produces ADP plus inorganic phosphate (Pi), which lowers the ATP/ADP ratio, a signal to increase ADP uptake into the mitochondria and increase ATP synthesis.

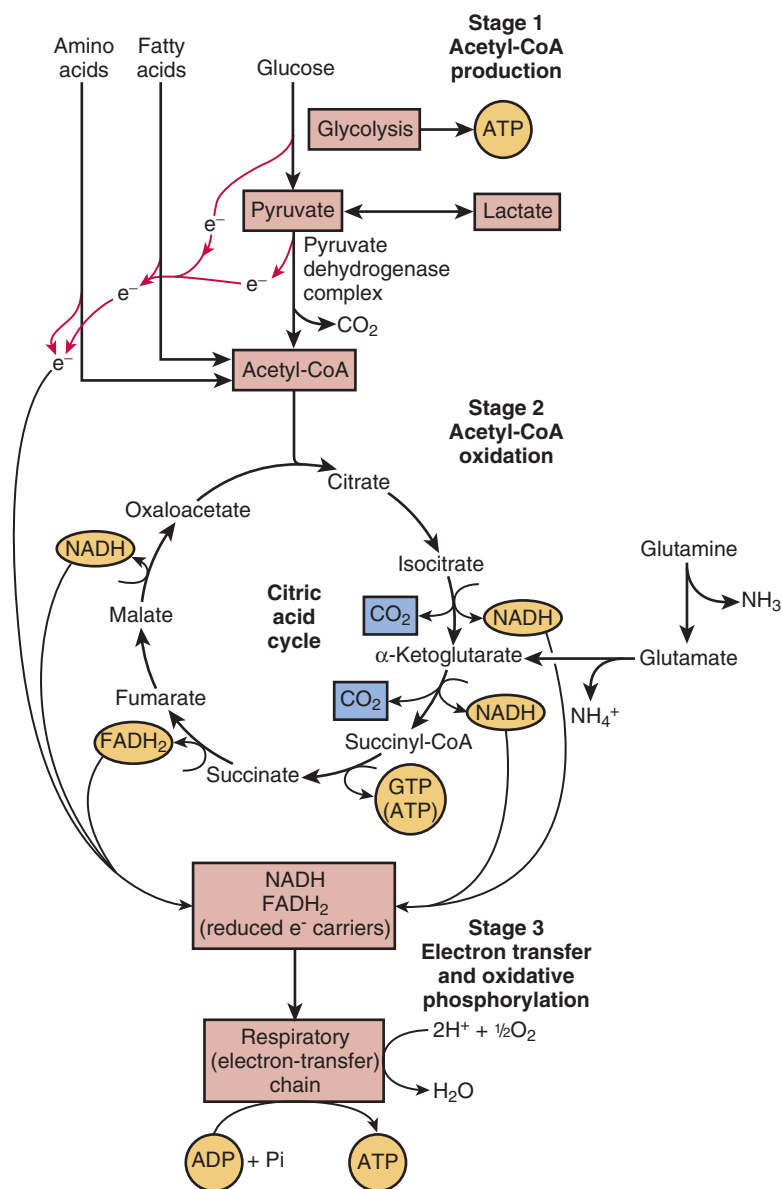


Fig. 5.4 Catabolism of proteins, fats, and carbohydrates in three stages of cellular respiration. Stage 1: oxidation of fatty acids, glucose, and some amino acids yields acetyl-coenzyme A (CoA). Stage 2: oxidation of acetyl groups in the citric acid cycle includes four steps in which electrons are abstracted. Stage 3: electrons carried by reduced nicotinamide adenine dinucleotide (NADH) and reduced flavin adenine dinucleotide (FADH_2) are funneled into a chain of mitochondrial (or, in bacteria, plasma membrane-bound) electron carriers (the respiratory chain) that ultimately reduces O_2 to H_2O . This electron flow drives the production of ATP. Also indicated are two proximal tubule pathways: (1) oxidation of lactate through pyruvate and acetyl-CoA, and (2) glutamine conversion to glutamate and α -ketoglutarate in the mitochondria with the production of two molecules of NH_3 , which is the main source of NH_3 secreted during acidosis. (Modified from Nelson DL, Cox MM. *Lehninger principles of biochemistry*, 5th ed. New York: WH Freeman; 2008.)

(NADH), one molecule of reduced flavin adenine dinucleotide (FADH_2), one molecule of guanosine triphosphate (GTP) or ATP, and two molecules of CO_2 are released in oxidative decarboxylation reactions (Table 5.1). Electrons carried by NADH and FADH_2 are transferred into the mitochondrial electron transport chain, a series of integral membrane complexes located within the inner mitochondrial membrane, where the electrons are sequentially transferred, ultimately to oxygen, which is reduced to H_2O . NADH and FADH_2 oxidation provoke the transport of H^+ from the matrix to the inner mitochondrial space.

The release of the potential energy stored in the H^+ gradient across the inner mitochondrial membrane provides the driving force for ATP synthesis from ADP by the ATP synthase: H^+ is transported into the matrix coupled to the production of ATP from ADP and inorganic phosphate (Pi) (see Fig. 5.3). These are the fundamental pieces of the chemiosmotic mechanism of oxidative phosphorylation proposed by Peter Mitchell in 1961.²² The newly synthesized ATP is extruded from the matrix into the intermembrane space via the ADP-ATP countertransporter known as adenine nucleotide translocase and then exits the mitochondria

Table 5.1 Adenosine Triphosphate (ATP) Yield From Metabolism of One Glucose Molecule

Process	Direct Product	Final ATP
ATP Yield from Complete Oxidation of Glucose		
Glycolysis	2 NADH (cytosol)	5 ^a
	2 ATP	2
Pyruvate oxidation (two per glucose)	2 NADH (mitochondrial matrix)	5
Acetyl-coenzyme A oxidation in citric acid cycle (two per glucose)	6 NADH (mitochondrial matrix)	18
	2 FADH ₂	4
Total yield per glucose		30
ATP Yield from Glycolysis of Glucose		
Glycolysis	2 ATP, 2 NADH	2

^aVia malate–aspartate shuttle.

across the permeable outer membrane. In the cytosol, ATP is available to bind to ATPases such as plasma membrane Na⁺-K⁺-ATPase.

In summary, the flow of electrons through the ETC generates a proton gradient across the inner mitochondrial membrane that provides the energy to drive ATP synthesis from ADP + Pi by ATP synthase and is also sufficient to extrude the ATP across the mitochondrial membrane.²³ Thus, the oxidation of substrates is coupled to ATP synthesis by an electrochemical proton gradient. This coupling can be influenced by uncoupling protein isoforms (UCPs) located in the mitochondrial inner membrane and expressed in a tissue-specific manner. Simply stated, UCPs create a proton leak that dissipates the proton gradient available to drive oxidative phosphorylation (see Fig. 5.3). It has been reported that UCP-2 is expressed in the renal proximal tubule and TAL (not in glomerulus or the distal nephron) and that its expression is elevated in kidneys of diabetic rats.²⁵ However, the physiologic consequences of the expression and regulation of UCP in kidneys has not been explored much experimentally.

WHITTAM MODEL

In the early 1960s, the coupling between active transport, respiration, and Na⁺-K⁺-ATPase activity was recognized by Whittam and Blond,^{69,70} who tested the idea that inhibition of active ion transport at the plasma membrane would cause a fall in oxygen consumption (QO₂) in the mitochondria. Using brain or kidney samples studied *in vitro*, they demonstrated that inhibition of Na⁺-K⁺-ATPase activity by removal of sodium or addition of the sodium pump–specific inhibitor ouabain (neither of which directly inhibits mitochondrial respiration) markedly reduced QO₂, which led the investigators to conclude that an extramitochondrial ATPase, sensitive to Na⁺ and ouabain, as well as to K⁺ and Ca⁺⁺, is one of the pacemakers of respiration of the kidney cortex.^{26,27}

A careful study by Balaban and colleagues two decades later²⁸ used a suspension of renal cortical tubules to reexamine this Whittam model (see Fig. 5.3) in more detail by measuring

the redox state of mitochondrial nicotinamide adenine dinucleotide (NAD), cellular ATP and ADP concentrations, ATP/ADP ratio, and QO₂ in the same samples. If transport and respiration are assumed to be coupled, inhibition of transport is predicted to provoke a mitochondrial transition to a resting state²⁹ accompanied by an increase in NADH/NAD⁺ (reduced to oxidized NAD), increase in [ATP], decrease in [ADP] and [Pi], increase in ATP/ADP ratio, and decrease in QO₂. Stimulation of active transport would provoke the opposite pattern: decreased NADH, ATP, and ATP/ADP ratio, and increased QO₂. Predictably, incubating the renal cortical tubule suspension with the Na⁺-K⁺-ATPase inhibitor ouabain caused a 50% decline in QO₂, reduction of NAD to NADH, and a 30% increase in the ATP/ADP ratio, all evidence for coupling of mitochondrial ATP production to ATP consumption via Na⁺-K⁺-ATPase. Similarly, in tubules deprived of K⁺ (which is required for Na⁺-K⁺-ATPase turnover), adding 5 mmol/L K⁺ increased QO₂ by more than 50%, oxidized NADH to NAD⁺, and decreased the cellular ATP/ADP ratio by 50%. These results provide evidence for the coupling of both Na⁺-K⁺-ATPase and ATP production via ATP synthase to the cellular ATP/ADP ratio (see Fig. 5.3).

ENERGY REQUIREMENTS AND SUBSTRATE USE ALONG THE NEPHRON

In all renal epithelial cells from the proximal convoluted tubule to the inner medullary collecting duct (IMCD), the basolateral sodium pump uses the hydrolysis of ATP to drive primary active transport of Na⁺ out of and K⁺ into the cell, and the gradients created are used to drive coupled transport of ions and substrates across both the apical and basolateral membranes.

Despite consistent distribution and function, the relative abundance of Na,K-ATPase as a function of tubular location along the nephron is highly variable. Na,K-ATPase activity, ouabain binding, and Na,K-ATPase subunit abundance have been studied in dissected tubules and with imaging techniques. Na,K-ATPase expression patterns and ouabain binding patterns along the nephron are very similar.^{10,30} The pronounced differences in activity can largely be accounted for by differences in sodium pump number measured either by ouabain binding or by immunoblot of subunits in dissected nephron segments (Fig. 5.5).³¹

The patterns of Na⁺-K⁺-ATPase protein expression and activity as a function of tubule length are what is to be expected from what is understood of the physiology of the nephron segments: moderate levels are expressed in the proximal tubule where two-thirds of the sodium is reabsorbed across a leaky epithelium, and lower levels are expressed in the straight than in the convoluted segments, reflecting the amount of sodium transported in these two regions. Very low levels are detected in the thin limbs of the loop of Henle, whereas high levels are expressed in the medullary and cortical TAL (“diluting segments”) that must reabsorb a significant fraction of NaCl without water against an increasingly steep transepithelial gradient. The Na⁺-K⁺-ATPase activity and expression in the DCT, which is responsible for reabsorbing another 5% to 7% of the filtered load against a very steep transepithelial gradient, is very high. In the collecting duct, which reabsorbs a smaller fraction of Na⁺ via channels electrically coupled to the secretion of K⁺ or H⁺ and has variable

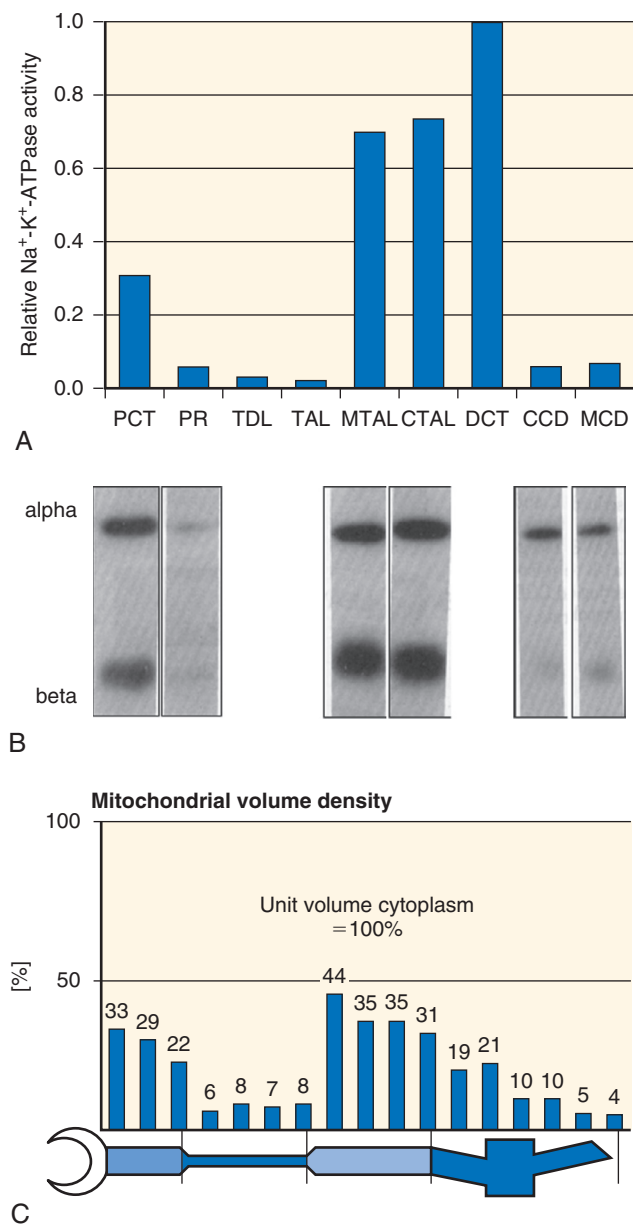


Fig. 5.5 (A) Relative levels of $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity measured in individual segments of the rat nephron. (Data are normalized to that of the distal convoluted tubule and expressed per unit length of tubule segment.) (B) Detection of $\text{Na}^+\text{-K}^+\text{-ATPase}$ α_1 - and β_1 -subunits along the nephron. Tubule segments 40 mm long were resolved by sodium dodecyl sulfate–polyacrylamide gel electrophoresis and subjected to immunoblotting with subunit-specific antisera. Blots placed below corresponding tubule label indicated in (A). (C) Morphologic analysis of mitochondrial density relative to a unit of cytoplasm. CCD, Cortical collecting duct; CTAL, cortical thick ascending limb of the loop of Henle; DCT, distal convoluted tubule; MCD, outer medullary collecting duct; MTAL, medullary thick ascending limb of the loop of Henle; PCT, proximal convoluted tubule; PR, pars recta (proximal straight tubule); TAL, thin ascending limb of the loop of Henle; TDL, thin descending limb of the loop of Henle. ([A] redrawn from Katz AI, Doucet A, Morel F. $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity along the rabbit, rat, and mouse nephron. *Am J Physiol.* 1979;237:F114–F120; [B] based on data from McDonough AA, Magyar CE, Komatsu Y. Expression of $\text{Na}^+\text{-K}^+\text{-ATPase}$ alpha- and beta-subunits along rat nephron: isoform specificity and response to hypokalemia. *Am J Physiol.* 1994;267:C901–C908; [C] based on data from Pfaller W, Rittinger M. Quantitative morphology of the rat kidney. *Int J Biochem.* 1980;12[1-2]:17–22.)

H_2O permeability, the $\text{Na}^+\text{-K}^+\text{-ATPase}$ is quite low, albeit sufficient to drive sodium reabsorption in this region. The distribution of the ATP-producing mitochondria along the nephron, reported as percent of cytoplasmic volume,³² parallels the distribution of the ATP-consuming sodium pumps but is somewhat less variable, ranging from 10% or less of the cell volume in the thin loop of Henle and medullary collecting duct to 20% in the cortical collecting duct (CCD) and proximal straight tubule to 30% to 40% of cell volume in the proximal tubule and TAL³² (Fig. 5.5C).

Determining which substrates support ATP production and $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity along the nephron has been the subject of many studies and reviews.^{24,33,34} To obtain nephron-specific information, investigators have dissected nephron segments and assayed for either metabolic pathway enzyme distribution or examined how specific substrates affected ATP levels. Although these in vitro approaches lack the in vivo realities of blood flow, tubular flow, and autocrine–paracrine, hormonal, and nervous system inputs that are evident in the whole kidney, the studies do provide information about the metabolic potential of each segment under defined conditions.

Isolated nephron segments had been reported to have low levels of cellular ATP, so Uchida and Endou³⁵ reasoned that if the segments were incubated with fuels that could be used by the segment, their ATP levels should increase toward physiologic levels. They examined a range of substrates for their ability to maintain cellular ATP levels in microdissected glomeruli and nephron segments (excluding thin sections of loop of Henle and papillary duct). The substrates studied (all at 2 mmol/L) included L-glutamine, D-glucose, β -hydroxybutyrate (HBA), and DL-lactate. Because the preincubation did not fully deplete the TAL and distal nephron segments of ATP, the ionophore monensin was included in the incubation with the substrate to dissipate the Na^+ gradient and promote ATP consumption.

The change in ATP per millimeter of tubule (or glomerulus) as a function of substrate addition, shown in Fig. 5.6, illustrates that each segment had a distinct ability to use these substrates. Lactate was very effective at maintaining ATP levels in all nephron segments tested, notably in the proximal tubule. The S1, S2, and S3 segments of the proximal tubule all used glutamine effectively as a fuel, which is consistent with the role of the proximal tubule in ammoniogenesis. Glutamine is the main amino acid oxidized by the proximal tubule, where it is deaminated and converted to $\alpha\text{-KG}$, yielding 2 NH_3 molecules that are secreted during acidosis, as illustrated in Fig. 5.4 and discussed in Chapter 9. Glutamine is not a preferred fuel in the more distal nephron segments. Glucose is completely reabsorbed along the proximal tubule yet, glucose is not an effective metabolic fuel for the S1 or S2 regions of the proximal tubule. In contrast, all of the more distal segments tested readily used glucose to maintain cellular ATP. The ketone HBA was used effectively in all nephron segments tested; however, in S1 and S2 of the proximal tubule the capacity of HBA to support ATP production was far less than that provided by glutamine or lactate.

The distribution along the nephron of numerous enzymes involved in metabolic pathways, collated from many studies, has been summarized by Guder and Ross.³³ Their description of glycolytic (Fig. 5.7A) and gluconeogenic (Fig. 5.7B) enzymes along the rat nephron^{36–38} demonstrates very low

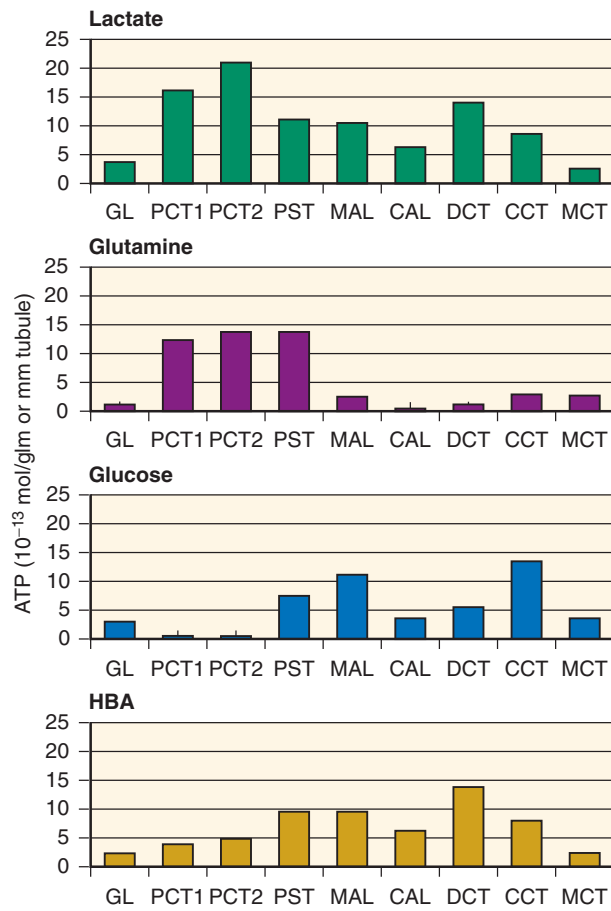


Fig. 5.6 ATP production in glomeruli and dissected nephron segments as a function of substrates. In glomeruli and PCT1, PCT2, and PST segments, the values equal the differences in ATP content between samples incubated with and without each substrate for 30 minutes. In MAL, CAL, DCT, CCT, and MCT, the values equal the differences in ATP content between samples incubated with and without each substrate in the presence of monensin (10 pg/mL) for 15 minutes. CAL, Cortical ascending limb; CCT, cortical collecting tubule; DCT, distal convoluted tubule; GL, glomerulus; MAL, medullary thick ascending limb; MCT, medullary collecting tubule; PCT1, early proximal convoluted tubule; PCT2, late proximal convoluted tubule; PST, proximal straight tubule. (Data from Uchida S, Endou H. Substrate specificity to maintain cellular ATP along the mouse nephron. *Am J Physiol.* 1988;255:F977–F983.)

glycolytic potential in the proximal tubule and high glycolytic potential from medullary ascending limb to medullary collecting tubule. In contrast, gluconeogenic enzymes are found almost exclusively in the proximal tubule.

In summary, the proximal tubule reabsorbs glucose and can synthesize glucose biosynthetically but does not metabolize glucose. There are both practical and theoretical explanations for the lack of glucose metabolism in this segment. The proximal tubule is specialized to reabsorb the filtered load of glucose from the tubular fluid back into the blood. Because of the enormous load of glucose moving through these cells, a proximal tubule hexokinase would need to have exceedingly low affinity for glucose, which would be difficult to regulate. In contrast, more distal regions of the nephron such as the loop of Henle and distal nephron normally have little or no

glucose in their tubular fluid, have no Na–glucose cotransporters in their apical membranes, and cannot synthesize glucose, but these regions use glucose delivered via RBF as a metabolic fuel (which could be provided by gluconeogenesis in the proximal tubule during fasting). A summary of substrate preferences along the nephron is provided in Fig. 5.8.³⁴

RENAL GLUCONEOGENESIS AND LACTATE HANDLING

In a review of renal gluconeogenesis, Gerich and colleagues³⁹ comment that the kidney can be considered two separate organs, because the proximal tubule makes and releases glucose from noncarbohydrate precursors, whereas glucose utilization occurs primarily in the medulla. Because the kidney is both a consumer and a producer of glucose, net arterio-venous glucose differences across the kidney can be uninformative, because glucose consumption in the medulla can mask glucose release by the cortex.

Gerich and colleagues³⁹ also make the case that the kidney is a significant gluconeogenic organ in normal humans based on the following: (1) in humans fasted overnight, proximal tubule gluconeogenesis can be as much as 40% of whole-body gluconeogenesis³⁹; (2) during liver transplantation, endogenous glucose release falls to only 50% of control levels by 1 hour after liver removal⁴⁰; and (3) pathologically in type 2 diabetes, renal glucose release is increased by about the same fraction as hepatic glucose release.⁴¹ Zucker diabetic fatty rats also exhibit marked stimulation of gluconeogenesis compared with their lean litter mate controls.⁴²

Lactate can reach the nephron by filtration or blood flow and can also be produced along the nephron. Within the kidney, lactate can be (1) oxidized to produce energy with generation of CO₂, a process that consumes oxygen but generates ATP; or (2) converted to glucose via gluconeogenesis in the proximal tubule, a process that consumes oxygen and ATP. This is shown in Fig. 5.9. Studies by Cohen⁴³ in an isolated whole kidney perfused with just lactate as substrate demonstrated a change in ¹⁴C–lactate utilization as a function of its concentration in the perfusate: at low concentrations, all the lactate was oxidized (detected as CO₂) in order to fuel transport and basal metabolism; when lactate in perfusate was raised above 2 mmol/L some of the lactate was used for synthesis of glucose (gluconeogenesis); and at high lactate in perfusate the metabolic and synthetic rates approach maximum, and some lactate is conserved (reabsorbed). However, it is not the normal circumstance that lactate is the sole substrate, and it is now appreciated that the metabolism of lactate is affected by the presence of other substrates—for example, lactate uptake and oxidation are inhibited in the presence of fatty acids.^{24,44}

The kidney's ability to convert lactate to glucose provides evidence that it can participate in cell–cell lactate shuttle, also known as the Cori cycle.⁴⁵ This cycle is important when oxidative phosphorylation is inhibited in vigorously exercising muscle, which becomes hypoxic. In the muscle, pyruvate is reduced to lactate to regenerate NAD⁺ from NADH, which is necessary for ATP production by glycolysis to continue. Lactate is released into the blood and can be taken up by tissues capable of gluconeogenesis, such as the liver and kidney. In the proximal tubule, the lactate that is not oxidized can be converted to glucose, and because this substrate is

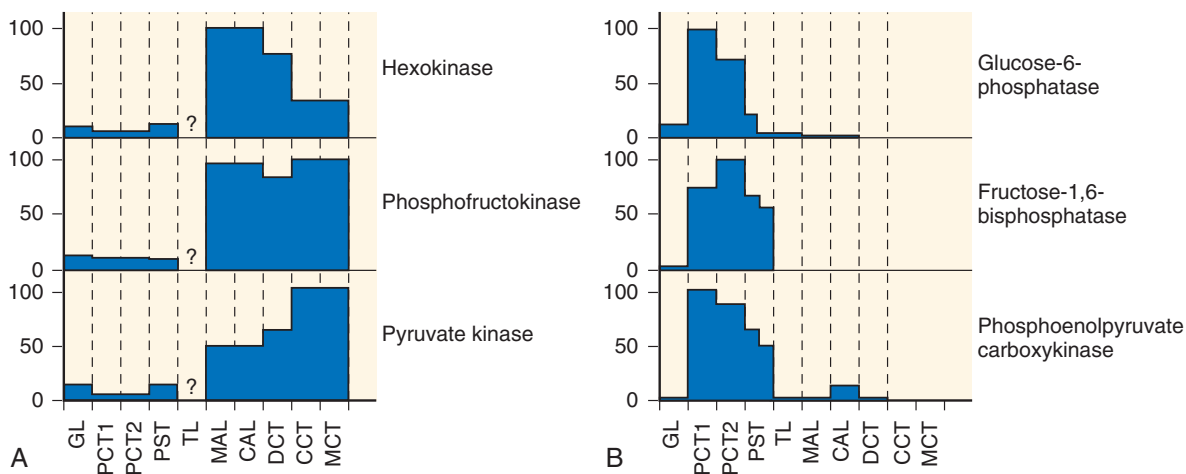


Fig. 5.7 Distribution of glycolytic and gluconeogenic enzymes along the rat nephron. Nephron segments were dissected from fed (A) and starved (B) rats, respectively. The activity of hexokinase, phosphofructokinase, pyruvate kinase, glucose-6-phosphatase, fructose 1,6-bisphosphatase, and phosphoenolpyruvate carboxykinase was determined in individual segments. Enzyme activities are expressed as a percentage of the maximal value observed, based on the original activity per gram of dry weight. CAL, Cortical ascending limb; CCT, cortical collecting tubule; DCT, distal convoluted tubule; GL, glomerulus; MAL, medullary thick ascending limb; MCT, medullary collecting tubule; PCT1, early proximal convoluted tubule; PCT2, late proximal convoluted tubule; PST, proximal straight tubule; TL, loop of Henle, thin limbs. (Modified from Guder WG, Ross BD. Enzyme distribution along the nephron. *Kidney Int.* 1984;26[2]:101–111.)

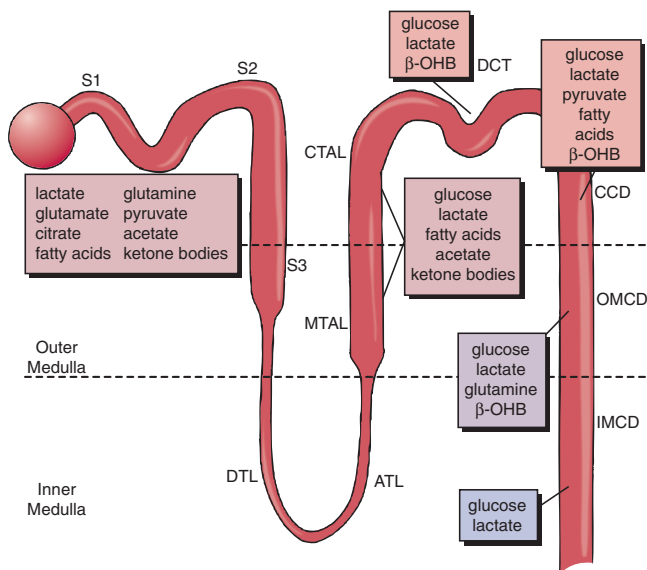


Fig. 5.8 Substrate preferences along the nephron. Summary of preferred substrates to fuel active transport in nephron segments as gleaned primarily from studies using oxygen consumption (Q_{O_2}), ion fluxes, radioactive carbon (^{14}C)-labeled carbon dioxide generation from ^{14}C -labeled substrates, ATP contents, and reduced nicotinamide adenine dinucleotide fluorescence. ATL, Ascending thin limb; β -OHB, β -Hydroxybutyrate; CCD, cortical collecting duct; CTAL, cortical thick ascending limb of the loop of Henle; DCT, distal convoluted tubule; DTL, descending thin limb; IMCD, inner medullary collecting duct; MTAL, medullary thick ascending limb of the loop of Henle; OMCD, outer medullary collecting duct; S1, S2, S3, successive segments of proximal tubule. (From Kone BC. Metabolic basis of solute transport. In *Brenner and Rector's the kidney*, 5th ed. St Louis: Saunders; 2008.)

not used by the proximal tubule, glucose will be reabsorbed back into the blood, where it will be available for metabolism by the exercising muscle. Overall, this cycle is metabolically costly: glycolysis produces 2 ATP molecules at a cost of 6 ATP molecules consumed in the gluconeogenesis. Thus, the

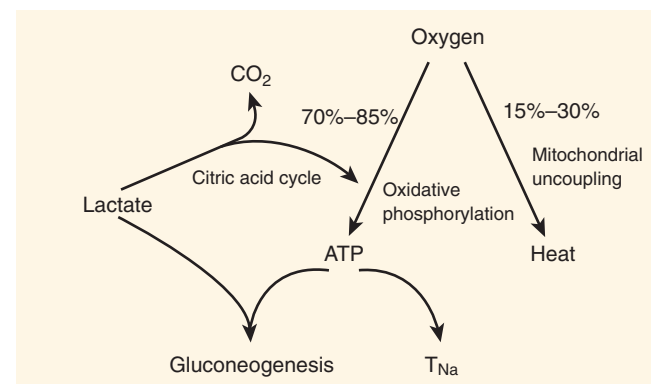


Fig. 5.9 Fate of lactate and oxygen in renal metabolism. Oxygen can be used to generate ATP via oxidative phosphorylation or heat if mitochondrial uncoupling occurs. Lactate can act as a substrate for gluconeogenesis, which consumes energy, or can enter into the citric acid cycle to generate energy. ATP is either used for Na transport (T_{Na}) or consumed in the process of gluconeogenesis.

Cori cycle is an energy-requiring process that shifts the metabolic burden away from the exercising muscle during hypoxia. This cell–cell lactate shuttle could also operate within the kidney between nephron segments that produce lactate anaerobically and the proximal tubule.

Renal medullary lactate concentration was explored in a 1965 study in rats by Scaglione and colleagues⁴⁶ to test the idea that the medulla used glycolysis in the low-oxygen environment. Medullary lactate concentration is a function of delivery via the blood flow, production in the medulla, and removal by the blood flow, because there is no gluconeogenesis in this region to consume lactate. Because of the countercurrent arrangement of the vasa recta, lactate would be expected to concentrate in the medulla somewhat. The study results indicated that lactate concentration was twice as high in the inner medulla as in the cortex and

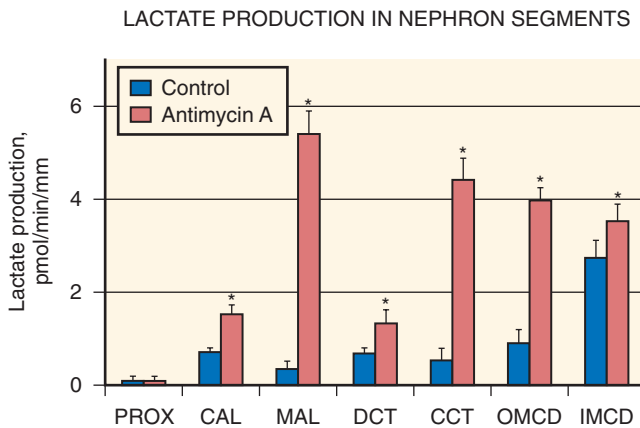


Fig. 5.10 Lactate production by rat nephron segments under control conditions and in the presence of antimycin A. CAL, Cortical ascending limb; CCT, cortical collecting tubule; DCT, distal convoluted tubule; IMCD, inner medullary collecting duct; MAL, medullary thick ascending limb; OMCD, outer medullary collecting duct; PROX, proximal tubule. (From Bagnasco S, Good D, Balaban R, et al. Lactate production in isolated segments of the rat nephron. *Am J Physiol.* 1985;248:F522–F526.)

that during osmotic diuresis the medullary lactate doubled, whereas cortical lactate remained unchanged. The authors postulated that increased medullary lactate was evidence for increased glycolysis during osmotic diuresis because the diuresis and increased flow through the vasa recta would be expected to decrease medullary lactate if synthesis rates were unchanged. Sodium delivery to the distal nephron would also increase during osmotic diuresis and the accompanying increased Na^+ reabsorption could drive the increased glycolysis.

Twenty years later, Bagnasco and colleagues⁴⁷ studied lactate production along the nephron in dissected rat nephron segments incubated in vitro with glucose with or without an inhibitor of oxidative metabolism, antimycin A. The only pathway for lactate production in the kidney is from pyruvate via lactate dehydrogenase. Proximal tubules produced no lactate with or without antimycin A. The distal segments all produced lactate, and the production was significantly increased (approximately 10-fold in TAL) during antimycin A incubation (Fig. 5.10), which led the authors to conclude that significant amounts of lactate can be produced by anaerobic glycolysis during anoxia in the distal segments. The IMCD, a region with low oxygen tension under control conditions, had high levels of lactate production even without antimycin A, which indicates that it is primed for anaerobic glycolysis.

NEPHRON-REGION-SPECIFIC METABOLIC CONSIDERATIONS

PROXIMAL TUBULE

Studies carried out in a number of laboratories provide evidence that sodium transport and gluconeogenesis compete for ATP in the proximal convoluted tubule.^{48–50} Friedrichs and Schoner⁴⁹ studied both processes in rat renal tubules

and slices and found that ouabain inhibition of $\text{Na}^+\text{-K}^+\text{-ATPase}$ increased renal gluconeogenesis by 10% to 40% depending on the substrate, and that stimulating $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity with high extracellular K^+ inhibited gluconeogenesis. The authors concluded that inhibition of the sodium pump induced a higher energy state of the cell, which would favor energy-requiring synthetic processes.

Nagami and Lee⁵⁰ used an isolated perfused mouse proximal tubule preparation to address this issue. When tubules were perfused at higher rates, delivering more sodium to the proximal tubule, the glucose production rate was decreased by 50%, whereas when tubules were incubated with ouabain in the bath or perfused with amiloride (to inhibit apical transport), the glucose production rate increased above that seen in nonperfused tubules. These authors also verified that the reduction in glucose production seen at elevated perfusion rates does not result from increased glucose utilization and is not dependent on the presence of specific substrates. Related to the topic of proximal tubule gluconeogenesis, two studies suggest that the blood glucose-lowering effects of sodium glucose cotransporter isoform 2 (SGLT2) inhibitors are due, in part, to reduced proximal gluconeogenesis: inhibiting SGLT2-mediated glucose uptake in diabetic mice reduced gluconeogenic gene expression, including PEPCK, a principal regulator of gluconeogenesis.^{51,52}

Recent studies and mathematical models have evaluated the effects of inhibiting proximal tubule transport routes on metabolism and transport efficiency (sodium transport/oxygen consumption, $T_{\text{Na}}/\text{QO}_2$, discussed further under the section regarding metabolic cost of sodium reabsorption). Proximal tubule transport, T_{Na} , is the sum of transcellular reabsorption (e.g., via Na^+/H^+ exchanger isoform 3 [NHE3]) and sodium glucose cotransporter isoform 2 (SGLT2), and paracellular reabsorption (e.g., via claudin 2 [cln2]). Mathematical models predict that $T_{\text{Na}}/\text{QO}_2$ is 80% higher in the S3 segment of the proximal tubule due to the larger paracellular contribution to T_{Na} , which is highly energy efficient. In these models, inhibiting NHE3 (the main sodium transporter in the proximal tubule), or Na,K-ATPase reduced T_{Na} , QO_2 , and transport efficiency.⁵³ Inhibiting SGLT2 also lowered T_{Na} but increased QO_2 in part by activating NHE3 and SGLT1 and in part by blunting the driving force for paracellular transport of T_{Na} due to tubular glucose buildup.^{51,53} Regarding paracellular reabsorption, QO_2 is markedly increased in proximal tubule claudin-2 null mice as T_{Na} is shifted from the energy efficient paracellular route to the Na,K-ATPase driven transcellular routes. Consequently, cln2 mice exhibit medullary hypoxia and increased susceptibility to tubular injury.^{54,55} How (or if) the increase in the metabolic cost of transport affects metabolic pathways and substrate utilization in the proximal tubule remains to be determined.

THICK ASCENDING LIMB

The TAL has a very high rate of Na^+ transport against a steep concentration gradient, very high levels of $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity and expression, and, perhaps not unexpectedly, 40% of its cytosolic volume occupied by mitochondria (see Fig. 5.5). Although the TALs have a far greater capacity for anaerobic metabolism than the proximal tubules, this region still requires oxidative metabolism to maintain cellular ATP levels and active Na^+ reabsorption.^{35,56}

CORTICAL COLLECTING DUCT

CCD metabolism is particularly interesting because it is made up of distinctly different cell types: principal cells that reabsorb sodium and intercalated cells that can secrete bicarbonate (HCO_3^-). Hering-Smith and Hamm microperfused rabbit CCD and measured Na^+ reabsorption (with $^{22}\text{Na}^+$) from lumen to bath, and HCO_3^- transport by microcalorimetry in the presence of substrates and with or without inhibitors. Both Na^+ reabsorption and HCO_3^- secretion were inhibited by antimycin A, which provides evidence for dependence on oxidative phosphorylation. However, neither was dependent on either glycolysis or the hexose–monophosphate shunt pathways. A small component of Na^+ transport was supported by endogenous substrates. Na^+ reabsorption was supported best by a mixture of basolateral glucose and acetate, whereas HCO_3^- secretion was fully supported by either glucose or acetate. HCO_3^- secretion (but not Na^+ transport) was supported to some extent by luminal glucose. In sum, this study indicates that principal cells and intercalated cells have distinct metabolic phenotypes.

MEDULLARY COLLECTING DUCT

Medullary collecting ducts contribute to final urinary acidification. Comparing the outer medullary collecting duct (OMCD) with the CCD, Hering-Smith and Hamm⁵⁷ found that bicarbonate secretion in the OMCD could be fully supported by endogenous substrates. This region has far less sodium transport and few mitochondria (see Fig. 5.5). Stokes and colleagues⁵⁸ isolated IMCDs and examined their metabolic characteristics. In the absence of exogenous substrate, IMCD can maintain cellular ATP and respire normally, which is evidence for the presence of significant endogenous substrate. In the presence of rotenone, an inhibitor of oxidative phosphorylation, glycolysis increased 56%, which provides evidence for anaerobic metabolism, as supported by enzymatic profiles. Inhibition of sodium pump activity reduced QO_2 by 25% to 35%, which provides evidence for a requirement

for a linkage between sodium pump activity and oxidative metabolism.

In studies that examined the metabolic determinants of K^+ transport in isolated IMCD,¹² glucose increased both oxygen consumption and cell K^+ content by more than 10%, whereas an inhibitor of glycolysis promoted a release of cell K^+ . Nor could cell K^+ content be maintained during inhibition of mitochondrial oxidative phosphorylation. Thus, in the IMCD, both glycolysis and oxidative phosphorylation are required to maintain optimal Na^+/K^+ -ATPase activity to preserve cellular K^+ gradients. Given the low PO_2 and low density of mitochondria in this region, the collecting ducts have a higher reliance on anaerobic metabolism, but still take advantage of oxidative metabolism to fully support transport.

SEXUAL DIMORPHIC PATTERN OF TRANSPORTERS ALONG THE NEPHRON AND METABOLIC CONSIDERATIONS

For 30 years, sex differences in renal hemodynamics, including lower GFR and higher RVR in females, and similar blood pressures between sexes, have been recognized in experimental rodents.⁵⁹ Immunoblots coupled to physiologic assays in rats indicate that females, versus males, exhibit a distinct transporter profile of lower proximal transporters' abundance and HCO_3^- reabsorption coupled to higher distal NCC and ENaC activation⁶⁰ (Fig. 5.11). This shift in T_{Na} from the energetically efficient PT to the costlier distal nephron is predicted to decrease sodium transport energy efficiency ($T_{\text{Na}}/\text{QO}_2$) in females. A rationale for this downstream shift in T_{Na} can be found in female biology. Pregnancy, and even more so lactation, represent major challenges to fluid homeostasis in females. The proximal tubule, which is shorter at baseline in females versus males,⁶¹ lengthens during lactation, driving a proportional increase in T_{Na} in a region where transport efficiency is very high due to paracellular T_{Na} .^{61,62} These significant sex- and reproduction-dependent differences in renal transport function likely necessitate differences in nephron region-specific metabolism that warrant serious future consideration.

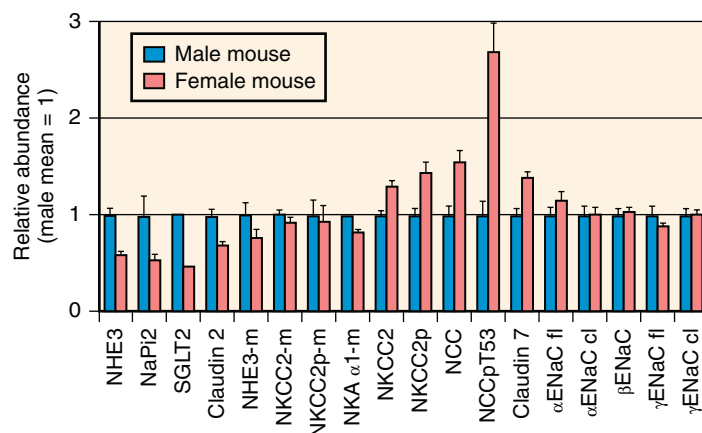


Fig. 5.11 Profile of renal sodium transporter protein abundance in female C57BL/6 mice expressed relative to abundance in male mice (defined as 1.0). Cortical and medullary (-m) samples were detected by immunoblot: Na^+/H^+ exchanger isoform 3 (NHE3), Na^+ -phosphate cotransporter isoform 2 (NaPi2), Na^+ -glucose cotransporter (SGLT2), claudin 2, $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ cotransporter isoform 2 (NKCC2 and activated phosphorylated form NKCC2p), Na,K -ATPase α catalytic isoform, Na^+/Cl^- cotransporter (NCC and activated phosphorylated form NCCp), claudin 7, epithelial Na^+ channel α,β,γ subunits full length (fl), and activated cleaved (cl) forms. (Adapted from Veiras et al.⁶⁰ and Pastor-Soler and Hallows KR¹⁹²).

CONTROL OF RENAL OXYGENATION

The kidneys are faced with the challenge of maintaining intrarenal oxygen levels so as to avoid both hypoxia, which leads to energy failure, and hyperoxia, which promotes oxidant damage.⁶³ Determinants of renal oxygenation and tissue oxygen tension (P_{O_2}) include (1) RBF and oxygen content of arterial blood; (2) the oxygen consumed by the cells; and (3) arterial-to-venous (AV) oxygen shunting, which entails the diffusion of oxygen from preglomerular arteries to postglomerular veins without being available to the cell for consumption.

RENAL BLOOD FLOW, OXYGEN CONSUMPTION, AND AV OXYGEN SHUNTING

The kidney enjoys a high blood flow, nearly 25% of the cardiac output, which is needed to sustain GFR. Compared with other major body organs, renal Q_{O_2} (product of RBF and renal oxygen extraction) per gram of tissue is high, second only to the heart (2.7 mmol/kg/min vs. 4.3 mmol/kg/min for the heart).¹ Renal Q_{O_2} is largely driven by the high RBF because renal oxygen extraction itself is low. Although RBF is high and renal oxygen extraction is low, the renal cortex is vulnerable to hypoxia.⁶⁴ It has been hypothesized that renal AV oxygen shunting is an adaptation to prevent hyperoxia in the setting of the high renal perfusion needed to sustain GFR. However, such shunting can be detrimental in conditions of oxygen demand-supply mismatch.⁶³

The phenomenon of O_2 shunting from descending to ascending vasa recta in the medulla has been accepted,⁶⁵ yet little else is known regarding the location of shunting in the cortex or its impact on oxygenation. Evidence for AV O_2 shunting in the kidney cortex was provided when it was shown, using oxygen-sensing microelectrodes, that the oxygen tension is substantially higher in the renal vein (50 mm Hg) than in efferent arterioles (45 mm Hg) or tubules (40 mm Hg).^{67,68} The fraction of oxygen subject to preglomerular AV O_2 shunting and impact on delivery of O_2 to the renal cortex was recently examined by independent groups using computer model simulations. A 2015 study concluded that preglomerular O_2 shunting was negligible and unlikely to impact renal oxygenation.⁶⁹ The issue was raised that the model may not have sufficiently considered the impact of wrapped artery–vein pairs (with short diffusion and no O_2 sink) that would facilitate AV O_2 shunting.⁷⁰ A 2017 study examining the preglomerular AV O_2 shunting concluded that although the AV shunting would have only a small impact under baseline conditions, it would exacerbate hypoxia during renal ischemia.⁷¹

Noting the similarity of tissue P_{O_2} in the kidney and in other organs, some have argued that the renal AV O_2 shunt is an adaptive mechanism for preventing the exposure of cortical tubules to toxic levels of oxygen while permitting a high RBF, which is needed for clearance.⁷² As mentioned earlier, there is substantial shunting of oxygen from descending to ascending vasa recta in the renal medulla due to countercurrent flow in these vessels. Countercurrent flow in “hairpin loops” formed by the vasa recta facilitates the recycling of solutes to the inner medulla, where a high osmolarity is essential to the formation of concentrated urine

(see Chapter 10). As an inherent consequence of this countercurrent mechanism for maintaining a medullary osmotic gradient, there arises a negative oxygen gradient from cortex to inner medulla, where P_{O_2} falls to 10 mm Hg.⁷³ This results from the combination of slow blood flow through the vasa recta, O_2 consumption by active transport in the outer medullary TAL, and diffusion of O_2 from descending to ascending vasa recta.⁷³ This leaves the medullary tissue at the brink of hypoxia, especially the stripe of outer medulla where the S3 segment of the proximal tubule and medullary TAL lie, making these segments most vulnerable to ischemic injury as they reabsorb significant fractions of filtered Na^+ .

Consideration of O_2 transport was incorporated into a mathematical model of the rat outer medulla by Chen and colleagues.^{74,75} The model takes into account fine details of the medullary anatomy, which includes positioning of the long descending vasa recta in the center of vascular bundles and the positioning of the TAL and collecting ducts at some distance from those vascular bundles. The model predicts steeply declining O_2 gradients from vascular bundles to the corresponding TALs and a compromise between the TAL and inner medulla with respect to the provision of oxygen.⁷⁶

In most organs, tissue oxygen can be stabilized by metabolic regulation of blood flow. In such an arrangement, vasoactive end products of metabolism due to increased metabolic activity and oxygen utilization produce a signal that results in more blood flow to that organ. A unique feature of renal oxygenation is that the kidney cannot rely on this simple mode of metabolic autoregulation because, unlike other organs that receive blood solely to supply the metabolic needs of the organ, the kidney also receives blood to perform the functions of glomerular filtration and tubule transport.

RBF creates its own demand because it determines GFR, which in turn determines the rate of sodium reabsorption, which is the main determinant of Q_{O_2} .^{77,78} If the kidneys were to modulate RBF as a means of stabilizing renal O_2 content, this would create a vicious cycle of positive feedback in which increased O_2 delivery would increase O_2 consumption, which would call for more O_2 delivery. Positive feedback is inherently destabilizing, so this arrangement alone could not work to stabilize either RBF or renal O_2 content. Hence, the kidney is compelled to invoke mechanisms that are more complex. There are two generic routes for the kidney to stabilize its O_2 content. One is to dissociate RBF from GFR. The other is to alter the metabolic efficiency of Na^+ transport (Table 5.2). Further details are discussed shortly.

Ultimately, the rate at which the kidney consumes oxygen must be linked to GFR. This is true because the main use

Table 5.2 Mechanisms for Changing the Amount of Oxygen Consumed per Work Performed

- Dissociate glomerular filtration rate from renal blood flow.
- Alter the amount of O_2 consumed per Na^+ reabsorbed.
- Shift transport between tubular segments that make more or less use of passive reabsorption.
- Alter back-leak permeability of the tubule.
- Change the coupling ratio of adenosine triphosphate generated to O_2 consumed by mitochondria.

of oxygen is to support the reabsorption of the filtered sodium, which is linked to GFR by glomerulotubular balance (GTB). GTB describes the direct effect of the filtered load on tubular reabsorption, and it operates in all nephron segments, although the mechanism differs between segments. In the proximal tubule, shear strain tied to increased tubular flow exerts torque on the apical microvilli, which leads to upregulation of apical sodium transporters.^{79,80} In cases in which filtration fraction increases, the parallel increase in peritubular capillary oncotic pressure will increase the Starling force driving fluid reabsorption. In the TAL, flux through NKCC2 is limited by chloride concentration, which declines more slowly along the TAL at high flow rates. But although GTB applies to net reabsorption, increased flow rate in the tubule also shortens the time that a given sodium ion is exposed to the reabsorptive machinery. This leads to the prediction that GTB can do no better than maintain constant fractional reabsorption.⁷⁸

TUBULOGLOMERULAR FEEDBACK

Significant fluctuations in RBF, GFR, and filtered Na^+ load would overwhelm the kidney's ability to accurately match Na^+ and volume output to input and compromise homeostasis of extracellular fluid volume. This does not normally occur because RBF and GFR are tightly controlled by the tubuloglomerular feedback (TGF) mechanism (described in detail in Chapter 3). In short, if RBF and/or GFR increases and GTB maintains a constant fractional reabsorption along the proximal tubule, an increasing amount of salt will be delivered to the macula densa, which sets off the TGF response. Specifically, increases in apical NaCl delivery or flow to this region provoke the cells of the macula densa to release ATP into the interstitium surrounding the afferent arterioles. This response is dependent on the basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$ to maintain the inward-directed Na^+ gradient.⁸¹ ATP release is via maxi-anion channels.⁸² Some fraction of the released ATP is converted to adenosine by local ecto-nucleoside triphosphate diphosphohydrolase 1 (ecto-NTPDase1) and ecto-5'-nucleotidase.⁸³ This adenosine activates A_1 adenosine receptors on the afferent arteriole, causing vasoconstriction. The arteriolar constriction reduces RBF and GFR in concert until Na^+ delivery to the macula densa is realigned. Thus, an inverse relationship is established between tubular NaCl load and the GFR of the same nephron.⁷⁷

Due to the time it takes for information to pass through the TGF system, the system is prone to oscillate with a period of around 30 seconds. Rhythmic oscillations of kidney Po_2 occur at the same frequency as TGF-mediated oscillations in tubular flow. This illustrates the simultaneous influence of TGF over minute-to-minute tubular flow rate and oxygen levels in the kidney.⁸⁴

Adenosine mediates TGF as a vasoconstrictor. Adenosine-mediated vasoconstriction is unique to the afferent arteriole. In all other beds where adenosine is vasoactive, it exerts a vasodilatory effect mediated by A_2 receptors. In addition to adenosine receptors, the afferent arteriole expresses $\text{P}_{2\text{X}}$ purinergic receptors that also mediate a vasoconstrictor response, in this case to interstitial ATP. These $\text{P}_{2\text{X}}$ receptors are essential to pressure-mediated RBF autoregulation,⁸⁵ but adenosine A_1 receptors are sufficient to explain the TGF response.⁸³

Adenosine also plays an important role in stabilizing medullary energy balance through local adjustments in blood flow and transport along with other autocrine and paracrine factors, including vasodilatory prostaglandins and nitric oxide, which increase medullary blood flow while inhibiting sodium transport in the TAL.^{48,86,87} Adenosine, in particular, is a case study in local metabolic regulation by negative feedback in the medulla. When ATP levels decline, adenosine is released from TAL cells into the renal interstitium, where it binds to adenosine A_1 receptors and inhibits Na^+ reabsorption in the TAL and IMCD. This has the effect of increasing Po_2 by reducing QO_2 . The same pool of adenosine also activates vascular adenosine A_2 receptors in the deep cortex and medullary vasa recta to increase blood flow.^{88,89}

By these mechanisms, the TAL looks after its own interest. But because TAL sodium reabsorption normally exceeds the urinary sodium excretion by 40-fold, any significant decline in TAL reabsorption must be compensated for by increasing active transport somewhere else or by reducing GFR through TGF. Activation of A_1 receptors in the glomerulus, proximal tubule, or TAL contributes to lessening the amount of work imposed on the hypoxic outer medulla, whereas activating A_2 receptors in the vasa recta supports O_2 delivery to the medulla (summarized in Fig. 5.12).

METABOLIC COST OF SODIUM REABSORPTION

The cost of renal sodium transport can be estimated from the sodium pump stoichiometry and the amount of oxygen required to produce ATP. Sodium pump stoichiometry dictates that hydrolysis of one ATP molecule is coupled to the transport of 3 Na^+ ions out of the cell and 2 K^+ ions into the cell,⁴ and oxidative metabolism generates approximately 6 ATP molecules per O_2 molecule consumed (see Table 5.1 and Fig. 5.4). In the 1960s, several investigators undertook measuring the metabolic cost of tubular reabsorption in various species of mammals. There is fair consensus among four oft-cited studies published between 1961 and 1966 that the relationship between QO_2 and T_{Na} is linear and that the kidney reabsorbs 25 to 29 Na^+ ions per molecule of O_2 consumed in the process.^{90–93} A representative figure from one of these studies is shown in Fig. 5.13.

If one assumes that kidney mitochondria make 6 molecules of ATP per molecule of O_2 , the kidney must then reabsorb 4 to 5 Na^+ per ATP molecule. This exceeds the 3:1 stoichiometry of the $\text{Na}^+\text{-K}^+\text{-ATPase}$, which was known at the time (reviewed in Burg and Good).⁹⁴ Because there are thermodynamic difficulties with the idea of an undiscovered basolateral sodium pump capable of forcing 5 Na^+ from a tubular cell with energy from a single ATP molecule, it was surmised that a considerable fraction of overall sodium reabsorption must be passive and paracellular, as is now accepted.

It was later suggested, by Cohen⁴³ and others, that these calculated ratios of $\text{QO}_2/\text{T}_{\text{Na}}$ actually underestimate the true efficiency of sodium reabsorption because a fraction of the oxygen consumed during Na^+ transport is also spent metabolizing organic substrates that enter the cell by Na cotransport. The most important example of this is lactate, which is converted to glucose in the proximal tubule via the Cori cycle. The capacity for renal gluconeogenesis from lactate is large, and it has been estimated that the kidney can consume

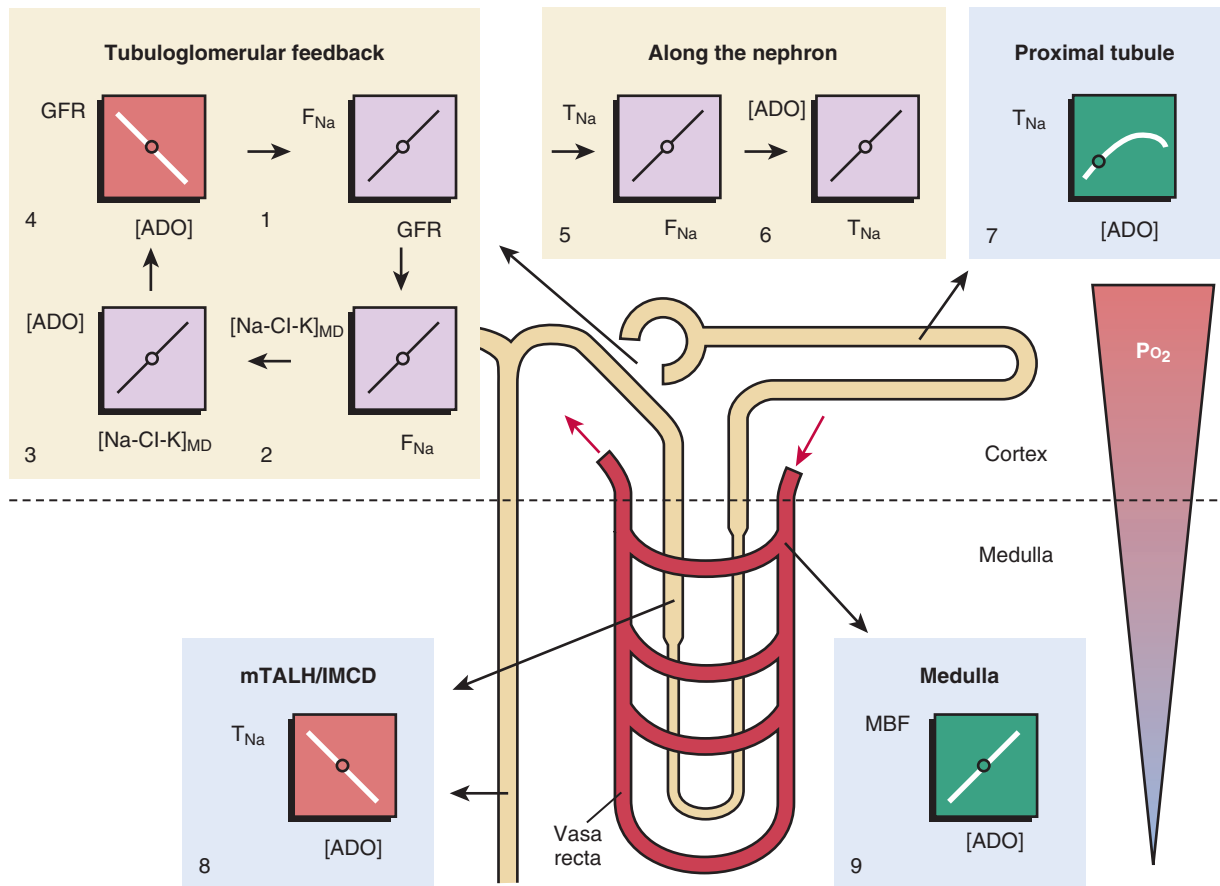


Fig. 5.12 Role of extracellular adenosine (ADO) in protecting the renal medulla from hypoxia. The line plots illustrate the relationships between the given parameters. *Small circles* on these lines indicate ambient physiologic conditions. (1) A rise in glomerular filtration rate (GFR) increases the Na^+ load (F_{Na}) to the tubular system in cortex and medulla. (2) This rise in F_{Na} increases the salt concentration sensed by the macula densa ($[Na-Cl-K]_{MD}$). (3) The increase in $[Na-Cl-K]_{MD}$, in turn, enhances local ADO. (4) ADO lowers GFR and thus F_{Na} , which closes a negative feedback loop and thus provides a basis for an oscillating system. (5) F_{Na} determines Na^+ transport work (T_{Na}) and O_2 consumption in every nephron segment, and thus oscillations in F_{Na} may help protect the medulla. (6) A rise in T_{Na} increases ADO along the nephron. (7) In the cortical proximal tubule, ADO stimulates T_{Na} and thus lowers the Na^+ load to segments residing in the medulla. (8) In contrast, ADO inhibits transport work in the medulla, including medullary thick ascending limb ($mTAL$) and inner medullary collecting duct ($IMCD$). (9) In addition, ADO enhances medullary blood flow (MBF), which increases O_2 delivery and further limits O_2 -consuming transport in the medulla. (Modified from Vallon V, et al. Adenosine and kidney function. *Physiol Rev.* 2006;86:901–940.)

up to 25% as much energy converting lactate to glucose as it spends reabsorbing sodium.⁴³

The metabolic cost of active sodium transport is expected to vary along the nephron. As reviewed earlier, the overall stoichiometry of Na^+ reabsorbed to O_2 consumed is estimated at 25 to 30 (microequivalents Na^+ /micromoles O_2).^{90,92} This ratio translates to 5 Na^+ reabsorbed for every ATP molecule consumed, which is much higher than the ratio of 3 Na^+ to every ATP molecule predicted by sodium pump stoichiometry. In fact, one might expect a ratio lower than 3 because of the basal metabolic functions of the kidney that are independent of sodium transport (i.e., insensitive to the Na^+-K^+ -ATPase inhibitor ouabain) (Fig. 5.14) and because of tubular back-leak.

One reason for this higher than expected efficiency of sodium reabsorption is that the kidney can leverage excess free energy in the gradients created by primary and secondary active transport to drive passive paracellular reabsorption of sodium chloride. The paracellular route in the proximal

tubule responsible for sodium reabsorption is the tight junction protein channel claudin-2.⁹⁵ Free energy for paracellular reabsorption is available in the mid-proximal tubule and early TAL.⁹⁴ In the proximal tubule, the driving force for the passive transport develops as a result of the preferential absorption of bicarbonate over chloride earlier in the tubule.^{96,97}

The decline in tubular bicarbonate concentration is paralleled by a rise in chloride concentration as water follows Na^+ , HCO_3^- , and organic osmoles across the leaky proximal tubule (see Chapter 5). This favorable lumen-to-blood Cl^- gradient drives passive paracellular chloride reabsorption. The transepithelial voltage that arises from electrodiffusion of chloride, in turn, drives passive sodium reabsorption. Because the $NaCl$ reflection coefficient is less than that for $NaHCO_3$ in this region,⁹⁶ coupled sodium chloride reabsorption also occurs secondary to solvent drag.⁹⁸ Although estimates vary, this passive reabsorption may increase the number of Na^+ ions reabsorbed to O_2 molecules consumed in the proximal tubule from 18 to 48.⁹⁹ As discussed in a previous section,

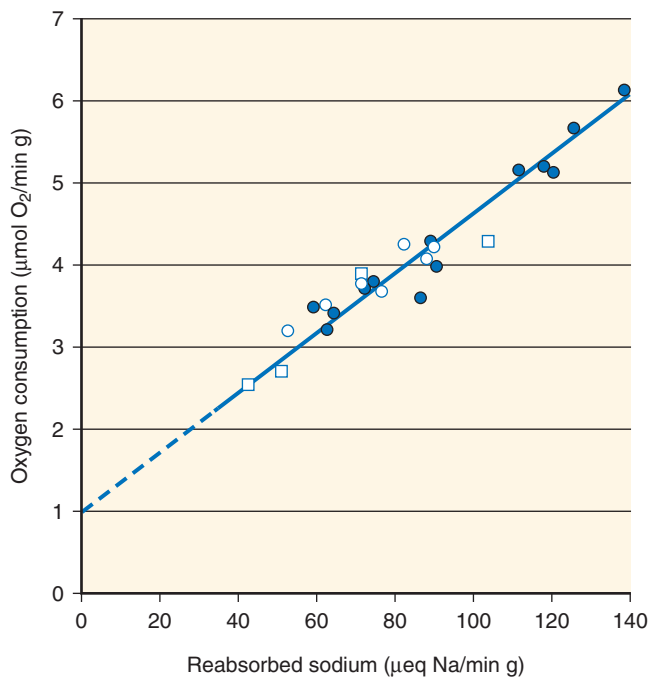


Fig. 5.13 Oxygen consumption as a function of net sodium reabsorption in whole dog kidney. Filled circles, control; open circles, hypoxia; squares, hydrochlorothiazide. The slope of the line fitted to the data points represents Q_{O_2}/T_{Na} and is approximately 1/28. (Modified from Thurnau K. Renal Na-reabsorption and O_2 -uptake in dogs during hypoxia and hydrochlorothiazide infusion. *Proc Soc Exp Biol Med*. 1961;106:714–717; and Mandel LJ, Balaban RS. Stoichiometry and coupling of active transport to oxidative metabolism in epithelial tissues. *Am J Physiol*. 1981;240[5]:F357–F371.)

Components of renal epithelial oxygen consumption

(Q _{O₂})	Ouabain sensitive • Primary active transport - $Na^+-K^+-ATPase$ • Coupled transport - Secondary Na-coupled transport - Tertiary coupled transport Ouabain insensitive (basal) • Primary and secondary active transport <i>not</i> coupled to $Na^+-K^+-ATPase$ • Cell repair, growth • Synthetic functions - Lipid synthesis - Gluconeogenesis • Substrate interconversions
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Fig. 5.14 A large fraction of renal epithelial oxygen consumption (Q_{O_2}) in renal cells is sensitive to the $Na^+-K^+-ATPase$ -specific inhibitor ouabain, and this Q_{O_2} drives primary active transport and transport coupled to sodium pump activity. The fraction of renal oxygen consumption that does not change in the presence of ouabain is, by definition, independent of $Na^+-K^+-ATPase$ activity in the cell and is roughly equivalent to the basal Q_{O_2} , which fuels transport not coupled to sodium gradients, cell repair and growth, biosynthesis, and substrate interconversions.

blunting paracellular reabsorption in the claudin-2-null mouse increases the oxygen consumption cost of T_{Na} by shifting it from paracellular to transcellular reabsorption, supporting the idea that paracellular transport is required for energy efficiency of oxygen utilization in the service of the large fraction of Na^+ reabsorption that occurs in the proximal tubule.⁵⁵

Simultaneously blocking both cytosolic and membrane carbonic anhydrase with acetazolamide reduces bicarbonate reabsorption and Q_{O_2} in a 16:1 molar ratio as expected for simple coupling to the sodium pump.¹⁰⁰ But inhibiting bicarbonate reabsorption with a membrane-specific carbonic anhydrase inhibitor, which acidifies the tubular lumen, paradoxically increases Q_{O_2} both in vivo and in isolated proximal tubules, an effect that is prevented by also blocking the apical Na^+/H^+ exchanger NHE3.¹⁰¹ A simple explanation is lacking for why increasing the cell-to-lumen proton gradient should increase Q_{O_2} in the proximal tubule, but these results establish the phenomenon in vitro and in vivo.

The early portion of the TAL is also capable of paracellular Na^+ reabsorption. In this region, Na^+ can be transported transcellularly by the apical $Na-K-2Cl$ cotransporter or apical Na^+/H^+ exchanger, secondary to a high density of $Na^+-K^+-ATPase$ extruding Na^+ across the basolateral membranes. In addition, Na^+ can be reabsorbed paracellularly as long as there is a lumen-positive transepithelial voltage sufficient to overcome the force for back diffusion associated with an unfavorable concentration difference. A lumen-positive voltage develops in the TAL because the apical membrane has a high concentration of K^+ channels, whereas the basolateral membrane has both K^+ and Cl^- channels. As predicted by the Goldman-Hodgkin-Katz voltage equation, the chloride conductance causes the basolateral membrane potential to be less negative than the apical membrane potential, which results in a positive transepithelial gradient.^{102,103}

Further along the nephron in the distal tubule and collecting duct, the tubular fluid sodium concentration is too low to allow paracellular reabsorption of sodium. In those segments, a lower limit on the cost of Na^+ reabsorption is set by the 3 $Na^+/1$ ATP ratio of the sodium pump. Although active transport of sodium is a pacemaker for renal respiration, there are ways to reset the relationship of Q_{O_2} to sodium pump activity. Examples of this were provided by Silva and Epstein, who measured both O_2 consumption and $Na^+-K^+-ATPase$ activity in rat kidney slices in which an increase in the latter had been induced by prior treatment of the animals with triiodothyronine (T_3), methylprednisolone, potassium loading, or subtotal nephrectomy. Although each of these maneuvers increased ex vivo sodium pump activity, only T_3 and methylprednisolone increased Q_{O_2} .¹⁰⁴

It has also been shown that the thermogenic effect of catecholamines, normally associated with brown fat and striated muscle, also occurs in the kidney, which responds to dopamine infusion with a near doubling of overall metabolic rate, but minimal change in sodium reabsorption.¹⁰⁵ Dopamine inhibits Na^+ reabsorption in the proximal tubule,^{106,107} thereby shifting the reabsorptive burden to less efficient downstream segments. However, heat accumulates in both cortex and medulla during dopamine infusion, which suggests that the mechanism may be a direct effect of catecholamines on renal metabolism. In addition, an increase in RBF

by dopamine could be responsible for the increased renal oxygen consumption, which is the product of RBF and renal oxygen extraction.

Weinstein and Szyjewicz^{108,109} also examined Q_{O_2}/T_{Na} using 10% body weight short-term saline expansion as another way to inhibit proximal Na^+ reabsorption in rats. This maneuver reduced fractional Na^+ reabsorption by 30% in the proximal tubule, leading to a GTB-mediated increase in net reabsorption downstream of the proximal tubule. Yet overall Q_{O_2} did not increase but actually fell. It was conjectured that energy for this increase in downstream reabsorption was derived anaerobically, but the full details of this remain to be clarified. It appears that the energy cost of transport in the proximal versus distal nephron during inhibition of proximal tubule transport depends on the nature of the stimulus provoking the change in transport as well as the metabolic environment.

Energetic efficiency of Na^+ reabsorption is highly variable along the nephron, in large part due to the presence of paracellular pathways for reabsorption in “leaky” epithelia such as the proximal tubule that can take advantage of favorable concentration and electrochemical gradients to reabsorb Na^+ without requiring ATP hydrolysis. Thus, if proximal reabsorption is lower (as in females) or reduced, the shift of T_{Na} to more distal sites should provoke an increase in Q_{O_2} driven by higher active uphill transcellular Na^+ reabsorption. The example of the *cldn-2* null mouse is provided earlier as an example.⁵⁵ Layton et al. now provide computational models that assess how shifting T_{Na} from one region to another or inhibiting T_{Na} in a region-specific manner alters overall Q_{O_2} .^{110,111} In this model, increasing single-nephron GFR did not alter the efficiency of reabsorbing Na^+ in the proximal tubule (T_{Na}/Q_{O_2}) while it increased T_{Na}/Q_{O_2} downstream secondary to higher paracellular T_{Na} due to higher luminal $[Na^+]$. This finding suggests that the proximal S3 segment and medullary thick ascending limb (which are vulnerable to hypoxia) are buffered from higher Q_{O_2} in response to increased flow and substrate delivery.¹¹⁰ When the model was applied to an examination of the effects of inhibiting individual transporters, the impact of inhibiting the proximal tubule NHE3 was most pronounced, not unexpected given the fact that the NHE3 reabsorbs 36% of filtered Na^+ at baseline. Overall, inhibiting NHE3 by 80% reduced GFR by 30% (mediated by TGF) and reduced whole kidney T_{Na}/Q_{O_2} by 20%, explained by far less proximal paracellular T_{Na} and the shift of T_{Na} to less efficient distal regions. Interestingly, the model predicts that NHE3 and NKCC2 inhibition increase CNT and CD ENaC-mediated T_{Na}/Q_{O_2} secondary to increased luminal Na^+ . Inhibition of NKCC2, NCC, or ENaC, although changing urine volume, Na^+ and K^+ excretion as predicted, did not change whole kidney T_{Na}/Q_{O_2} .¹¹¹

RENAL OXYGENATION AND METABOLISM DURING NORMAL PERTURBATIONS AND DISEASE

The kidneys have developed multiple mechanisms to minimize changes in oxygen delivery and to cope with a reduction in PO_2 . Some of these are specific to the kidney, whereas others are generic to many tissues as discussed next.

PHYSIOLOGIC REGULATION: FILTRATION FRACTION AND Q_{O_2}/T_{Na}

As reviewed earlier, there are two generic routes for the kidney to achieve a stable content of O_2 : dissociation of RBF from GFR, and alteration of Q_{O_2}/T_{Na} (see Table 5.2). Both routes are subject to regulation. Dissociating RBF from GFR equates to changing the glomerular filtration fraction. This can work to stabilize kidney O_2 because lowering the filtration fraction increases the ratio of supply to demand for O_2 . For nephrons in filtration equilibrium, this requires independent control of the afferent and efferent arterioles, which can be achieved by modulating relative activities of purinergic, angiotensin, nitric oxide, and other signaling systems in the glomerulus. A full discussion of glomerular hemodynamics is available in Chapter 3, but a few features are noted here.

To begin, filtration fraction can be lowered by constricting the afferent arteriole (which reduces net O_2 delivery) or dilating the efferent arteriole (which increases net O_2 delivery). Constricting the afferent arteriole confers initial energy savings by reducing GFR faster than RBF, but there is diminishing return as O_2 delivery declines toward the basal requirement. Dilating the efferent arteriole reduces GFR only when glomerular capillary pressure is low to begin with, such as during hypotension or with high afferent resistance.¹¹² When angiotensin II acts on the afferent and efferent arterioles to stabilize GFR in the face of low blood pressure or high upstream resistance by preferentially constricting efferent arterioles, the kidney is accepting a decrease in the ratio of O_2 supply to O_2 demand. Conversely, adenosine signaling in the glomerulus decreases filtration fraction and so manages to stabilize nephron function without compromising the O_2 supply–demand balance. Adenosine in the nanomolar range constricts the afferent arteriole via high-affinity A_1 adenosine receptors. Higher adenosine concentration dilates the efferent arteriole via low-affinity A_2 adenosine receptors. Interstitial adenosine concentration rises as more NaCl is delivered into the nephron. The prototype for this is the TGF signaling through the macula densa, although other sources are not precluded (see Fig. 5.12). When the kidney is operating in the domain of modest distal delivery, increasing the TGF signal constricts the afferent arteriole. When the kidney is operating in the domain of high distal delivery, further increase causes the efferent arteriole to dilate,¹¹³ which can be viewed as a shift in priority toward maintaining the O_2 supply as the supply diminishes.

The second generic means for stabilizing kidney O_2 is to alter Q_{O_2}/T_{Na} . As mentioned earlier, studies in the 1960s established the linear relationship between Q_{O_2} and T_{Na} . Each adopted a similar standard, which was to express suprabasal renal Q_{O_2} as a function of T_{Na} . Suprabasal O_2 consumption was obtained by subtracting from total O_2 consumption the amount required for basal metabolism. The latter was determined by various methods. One method was to plot Q_{O_2} against T_{Na} and then extrapolate to the y-intercept to obtain basal Q_{O_2} . Another approach was to reduce renal perfusion pressure to the point that glomerular filtration ceased, then ascribe the residual measured Q_{O_2} to basal metabolism. These approaches for obtaining basal O_2 consumption have their unique limitations, and both require the dubious assumption that basal metabolism is static under most conditions and is unaffected by T_{Na} per se. Nonetheless,

these studies attempted to measure and incorporate the contribution of basal Q_{O_2} to the total Q_{O_2} . In the recent literature, the ratio of total Q_{O_2} to T_{Na} has been represented as an index of metabolic efficiency of transport, ignoring the contribution of basal Q_{O_2} to the total Q_{O_2} . This can lead to inaccuracies as estimates of basal metabolism have varied widely in the published literature, indicating its susceptibility to different experimental conditions.¹¹⁴ For example, the proximal tubule can devote considerable energy to gluconeogenesis, especially in the postabsorptive or fasting states, and in diabetes.^{41,115,116} In light of the fact that oxygen can be diverted to do other work, an increase in Q_{O_2}/T_{Na} is not necessarily due to “decreased transport efficiency.”

The specific factors contributing to this Q_{O_2}/T_{Na} stoichiometry as well as to the basal metabolic rate in the kidney have been the subject of numerous reviews.^{117,118} It is theoretically possible to alter Q_{O_2}/T_{Na} in a number of ways (see Table 5.2):

1. Transport could be shifted from the proximal tubule, where efficient use of energy from the Na^+K^+ -ATPase drives passive transport, to other segments where all Na^+ reabsorbed passes through the Na^+K^+ -ATPase.
2. Tubular back-leak permeability could change, which would affect the number of times that a given Na^+ ion must be reabsorbed to escape excretion into the urine.
3. The ratio of ATP produced per O_2 consumed could be altered by the regulated activity of UCPs (see Fig. 5.3).²⁵
4. ATP could be diverted to gluconeogenesis, such as during fasting.

The same neurohumoral factors that exert well-known effects on glomerular hemodynamics and O_2 supply, including nitric oxide, angiotensin II, adenosine, and catecholamines, also appear to participate in the regulation of kidney metabolism and Q_{O_2} by the tubule. It has been shown^{119,120} that the administration of nonselective nitric oxide synthase (NOS) inhibitors increases Q_{O_2}/T_{Na} . Other experiments suggested that NOS-1 is the specific isoform that regulates this action in vivo.¹¹⁹ The changes in Q_{O_2} with NOS inhibition may occur due to (1) a shift in the site of sodium reabsorption to a less efficient nephron segment; (2) decreased efficiency of transport in the proximal tubule (i.e., decrease in the passive component of reabsorption); or (3) less efficient use of O_2 by mitochondria. For example, nitric oxide given directly to a proximal tubular cell is both a proximal diuretic¹²¹ and a competitive inhibitor of O_2 flux through the ETC in mitochondria.¹²²

Most effects of nitric oxide are mediated by cyclic guanosine monophosphate, but the mitochondrial effect is presumed to occur through the competitive inhibition of cytochrome *c* oxidase.^{122–124} Studies in normal rats,¹²⁵ in rats with experimental diabetes,¹²⁶ and in rats with untreated hypertension^{127,128} have found an antagonistic relationship between nitric oxide and angiotensin II in terms of both glomerular hemodynamics and tubular reabsorption. Specifically, systemic NOS blockade causes renal vasoconstriction and activation of TGF, which can be prevented by angiotensin II blockers.

A similar antagonistic relationship appears to exist in control of kidney metabolism as well. Angiotensin II was recently shown to be capable of increasing Q_{O_2} despite lowering T_{Na} .¹²⁹ Rats and mice with angiotensin-induced hypertension exhibit stimulation of sodium transporters from

the cortical TAL to the CD, in regions with higher Q_{O_2}/T_{Na} , and inhibition or no stimulation of sodium transporters in the proximal nephron, where Q_{O_2}/T_{Na} is lower.^{130,131} Studies in spontaneous hypertensive rats have suggested opposing effects of angiotensin II and nitric oxide on the Q_{O_2}/T_{Na} ratio in the kidney.¹³² Rats with angiotensin-induced hypertension demonstrated an increased Q_{O_2}/T_{Na} that was reversed by a mimetic of superoxide dismutase, which is consistent with the theory that many angiotensin II effects are mediated by upregulating the activity of reduced nicotinamide adenine dinucleotide phosphate (NADPH) oxidase.¹²⁹ In addition, there is evidence that angiotensin II contributes to mitochondrial dysfunction and oxygen consumption in aging rats.¹³³

There is also evidence for a self-contained, intrarenal renin-angiotensin system that operates independently of the systemic renin-angiotensin system (RAS).^{134–137} It is possible to dissociate tubular and whole-kidney angiotensin II in the regulation of proximal reabsorption and salt homeostasis.¹³⁷ For example, a low-salt diet activates the systemic RAS and increases renal sodium reabsorption without any measurable increase in intrarenal synthesis of angiotensin II,¹³⁸ whereas a high-salt diet has a predictable inhibitory effect on plasma and whole-kidney angiotensin II but, surprisingly, leads to increased angiotensin II content of proximal tubular fluid. This finding explains why the tonic influence of endogenous angiotensin II over proximal reabsorption fails to decline with consumption of a high-salt diet. Thus, whereas the systemic RAS is oriented toward salt homeostasis, it appears that the tubular angiotensin II system is oriented toward a stable salt delivery beyond the proximal tubule.¹³⁷

The role of angiotensin II in kidney metabolism is implicated in the ablation/infarction remnant kidney model of chronic kidney disease (CKD). Oxygen consumption factored for nephron number or T_{Na} has been shown to be elevated in this model^{139–141} and lowered by various treatments including angiotensin blockade.¹³⁹ A connection has also been established recently between local accumulation of the Krebs cycle intermediate succinate and activation of the RAS.¹⁴² Succinate can accumulate extracellularly when oxygen supply does not match demand. In the extracellular fluid, it can bind to its G protein-coupled receptor, GPR91. PO_2 in the juxtaglomerular region is reduced in the hyperglycemia of diabetes, and succinate levels are very high in urine and renal tissue of diabetic animals. Inhibition of the Krebs cycle's succinate dehydrogenase complex causes robust renin release. This effect is amplified in high-glucose conditions or with added succinate. In summary, GPR91-mediated signaling in the juxtaglomerular apparatus could modulate glomerular filtration rate and RAS activity in response to changes in metabolism (especially after a meal when glucose level is elevated). Pathologically, GPR91-mediated signaling could link metabolic diseases (such as diabetes) with RAS activation, systemic hypertension, and organ injury.

HYPOXIA AND ISCHEMIA

Intrarenal hypoxia has been proposed as a final common pathway to progression of CKD.¹⁴³ In late stages of CKD, rarefaction of capillaries and other structural changes have been implicated in the decrease in oxygen supply leading to hypoxia. However, intrarenal hypoxia has been demonstrated in the early stages before any structural changes.¹⁴⁴ High

QO_2/T_{Na} has been postulated to be the etiology of tubular hypoxia in the early stages of CKD.¹³⁹ In early experimental diabetes, decreased tissue oxygen tension (PO_2) has also been demonstrated prior to any structural changes associated with diabetic nephropathy.¹⁴⁵ Various forms of kidney injury, acute and chronic, have demonstrated tissue hypoxia in early and late stages as discussed later in the chapter.

Blood oxygen level-dependent magnetic resonance imaging (BOLD MRI) has been used to measure blood flow, oxygen tension, and regional tissue oxygenation in kidney cortex and medulla in humans with hypertension. The following were compared: kidneys with atherosclerotic renal artery stenosis, the kidneys contralateral with the stenotic kidneys, and kidneys in individuals with essential hypertension with no accompanying stenosis.^{146,147} In the stenotic kidneys, as expected, tissue volume was decreased and blood flow was compromised; however, there was no significant decrease in PO_2 in the cortex or deep medulla compared with the contralateral kidney in the same person or compared with kidneys in individuals with essential hypertension. This led the authors to postulate that there was reduced oxygen consumption in the stenotic kidneys. Consistent with this interpretation, furosemide-suppressible QO_2 in the medulla was significantly less in the stenotic kidney than in the contralateral kidney or kidneys in individuals with essential hypertension.¹⁴⁷ The association between tissue hypoxia and renal damage has also been assessed in patients with renal artery stenosis.¹⁴⁸

In diabetic and nondiabetic CKD patients, intrarenal hypoxia correlates with renal pathology.^{149,150} Alterations in renal oxygenation in acute transplant rejection have also been described.¹⁵¹ Recently, the effect of sodium intake on renal tissue oxygenation was investigated.¹⁵² In brief, 1 week of low- Na^+ intake increased renal medullary oxygenation in both normotensive and hypertensive subjects, whereas a high- Na^+ diet reduced medullary oxygenation. Another recent study in patients with CKD and hypertension compared with healthy controls revealed tight regulation of renal oxygenation at rest but altered response to furosemide in both CKD and hypertension groups, suggesting early metabolic changes in hypertension.^{153,154} In many of these studies, furosemide was administered to inhibit tubular reabsorption, which improved medullary oxygenation, demonstrating the significant role of Na reabsorption-driven QO_2 , even in established disease with structural alterations, which impact oxygen delivery.

Assessment of tissue hypoxia in acute kidney injury has largely been described in animal studies. Studies in an ischemia-reperfusion model in rats and pigs demonstrated reduced oxygenation and persistent tissue hypoxia in the early stages after reperfusion,^{155,156} which was more prominent in the outer medullary regions. These studies were limited to early stages after reperfusion (3–4 hours post ischemia). Assessment of tissue oxygenation in later stages could provide important information regarding the role of hypoxia in repair or recovery from AKI and/or transition to CKD. In animal studies of sepsis associated AKI, variability in tissue oxygenation has been reported depending on the animal model, species, or time point after injury.^{157–161} However, increased renal oxygen extraction despite reduction in GFR and filtered load has been described,^{160,161} indicating inefficiency of oxygen utilization and/or changes in basal metabolism of tubular cells. Clinical studies assessing renal oxygenation in AKI are

limited. However, in a set of elegant studies in postcardiac surgery patients with or without AKI, increased renal O_2 extraction and QO_2/T_{Na} was demonstrated in the AKI group.¹⁶² Improvement in renal oxygenation with loop diuretics due to reduction in tubular transport related QO_2 was also demonstrated.¹⁶³

HYPOXIA-INDUCIBLE FACTOR

One of the major transcription factors mediating the cellular adaptation to hypoxia is the oxygen-sensitive, hypoxia-inducible factor (HIF).¹⁶⁴ A great body of work by Semenza have described the role of HIF as a primary oxygen sensor and regulator of cellular oxygen homeostasis.^{164–167} It accumulates in hypoxic cells, where it acts to regulate gene expression. HIF consists of a labile α subunit (HIF1 α , HIF2 α , HIF3 α) and a constitutive β subunit. These subunits heterodimerize to form a transcriptional complex that translocates to the nucleus and binds to hypoxia response elements of various hypoxia responsive genes.^{168,169} HIF1 α and HIF2 α have been well studied, have similar structures, and have significant overlap in their actions on target genes; however, some target genes appear to be exclusively under the regulation of one or the other. Their renal tissue expression patterns also differ, with predominant expression of HIF1 α in the tubular epithelial cells and HIF2 α in the interstitial fibroblasts and peritubular endothelial cells in the hypoxic kidney.^{170,171} There is limited information regarding the function and actions of HIF-3 α .

During adaptation to hypoxia, HIF1 α and 2 α regulate the expression of many genes that regulate oxygen delivery and consumption. Changes that culminate in a rise in erythropoiesis, vasodilation, and tissue vascularization all increase oxygen delivery.^{172,173} Another set of responses conserve energy by decreasing substrate movement into the tricarboxylic acid cycle, increasing cellular glucose uptake and glycolytic enzymes to increase anaerobic ATP production and shifting the cell towards glycolytic metabolism.¹⁶⁷ HIF-1 α also has significant effects on mitochondrial metabolism; specifically, it diminishes NADH supply to the ETC, induces a subunit switch in complex IV of ETC to optimize its efficiency in hypoxia, and represses mitochondrial biogenesis and respiration.^{174,175} Finally, HIF-1 α also induces mitochondrial autophagy as an adaptive metabolic response to prevent increased levels of reactive oxygen species (ROS) generation and cell death in hypoxia.¹⁷⁶

Recently, HIF-1 α has also been recognized as a regulator of salt transport. High-salt intake increased HIF-1 α expression in the renal medulla.¹⁷⁷ Inhibition of HIF-1 α in the renal medulla decreased medullary blood flow, and blunted urine flow and urinary Na excretion. In the presence of HIF-1 α inhibitor, rats on a high-salt intake developed positive cumulative salt balance and higher blood pressure. Thus, medullary HIF-1 α inhibition on a high-salt intake leads to resetting of pressure natriuresis, Na retention, and salt-sensitive hypertension.¹⁷⁸ HIF-1 α expression has been reported in the medulla in a normal rat,¹⁷⁹ where selective inhibition of medullary HIF-1 induced significant tubulointerstitial damage. Interestingly, HIF-1 expression appeared to correlate with salt transport—an increase noted after an increase in medullary workload and a decrease in expression noted after inhibiting Na transport in the TAL by furosemide administration, which also increased medullary PO_2 .¹⁷⁹

HIF activity is regulated by proteasomal degradation during normoxia by von Hippel-Lindau-E3 ubiquitin ligase complex after being hydroxylated by prolyl 4 hydroxylase domains (PHD). Of the three main identified PHDs (1, 2, and 3), PHD2 is the main enzyme that targets HIF for degradation under normoxia.¹⁷⁰ All three PHDs are expressed in the kidney and are found predominantly in the distal convoluted tubule, collecting duct, and podocytes, and levels are depressed during ischemia and reperfusion.^{169,170} HIF activity is also regulated by factor inhibiting HIF (FIH), which inhibits its transcriptional activity by hydroxylating an asparagine residue within the transactivation domain and prevents the binding of coactivators to the HIF transcriptional complex.¹⁶⁶

A tremendous amount of progress has been made in understanding how HIF helps to maintain O₂ homeostasis, and the study of this factor in the kidney under normal physiologic and pathophysiologic conditions. Particularly, the role of HIF stabilization and activation in the kidney to stimulate erythropoietin production has been targeted therapeutically. Structural analogs of 2-oxyglutarate, which are the substrates used by PHDs for HIF hydroxylation, have been developed to reversibly inhibit PHD activity. These analogs are in phase 2 and 3 clinical trials for treatment of anemia by endogenous erythropoietin production in CKD.¹⁸⁰ Recently, the efficacy of Vadadustat, an oral PHD inhibitor, was evaluated in a 20-week, double-blind, randomized, placebo-controlled, phase IIb study in patients with stage 3–5 CKD.¹⁸¹ Hemoglobin levels were increased and maintained in the treatment arm without major adverse effects. Moreover, increased iron mobilization was also reported in patients treated with Vadadustat. Another oral PHD inhibitor, Roxadustat, was evaluated in an open-label, phase IIb study in patients with CKD using different starting doses and frequencies of administration for 16 or 24 weeks.¹⁸² The various dosing regimens were well tolerated and effective in achieving target hemoglobin levels, without the need for additional iron supplementation. The same agent has also shown effectiveness in patients on incident hemodialysis or peritoneal dialysis.¹⁸³ A third agent, GSK1278863, was evaluated in both CKD patients not on dialysis and ESRD patients on dialysis in a 4-week, phase IIa study.¹⁸⁴ It was safe and well tolerated in both populations and produced dose-dependent increases in hemoglobin concentrations in the CKD patients, whereas only the higher dose was effective in maintaining hemoglobin concentrations in the ESRD patients on dialysis. The ease of administration, lower peak plasma erythropoietin levels, and beneficial effects on iron homeostasis make these oral HIF stabilizers attractive alternatives for anemia management in CKD. However, longer-duration trials to study efficacy and safety of these agents in CKD and ESRD patients are needed.

ADENOSINE MONOPHOSPHATE-ACTIVATED PROTEIN KINASE

The energy status of the cell can be detected by the ultrasensitive 5'-adenosine monophosphate (AMP)-activated protein kinase (AMPK), which is a ubiquitously expressed, highly conserved, key energy sensor and regulator of cellular metabolic activity.¹⁸⁵ AMPK consists of a heterotrimer of a catalytic subunit (α 1 or α 2), together with a beta (β 1 or β 2) and a gamma (γ 1, γ 2, or γ 3) regulatory subunit.¹⁸⁵ Cellular energy stress, which can be due to a variety of conditions, such as

nutrient or glucose deprivation, exercise, hypoxia, or ischemia, is detected as a rising concentration of AMP and an increase in the AMP/ATP ratio. AMPK is activated by phosphorylation of the α catalytic subunit on threonine-172 (Thr172) by upstream kinases.¹⁸⁶ The binding of AMP to the γ -regulatory subunit of AMPK increases its activity in three ways: (1) conformational change in AMPK, which allows enhanced phosphorylation of the α catalytic subunit on Thr172 by upstream kinases, thus, activating AMPK; (2) inhibition of dephosphorylation of the catalytic subunit; and (3) direct allosteric activation. These three effects, working in concert, render the system exquisitely sensitive to changes in AMP, and all are antagonized by ATP; thus the importance of the AMP/ATP ratio. AMPK acts as a metabolic checkpoint to facilitate metabolic adaptation to cellular energetic stress by triggering ATP producing pathways such as fatty acid oxidation, glucose uptake, and glycolysis, while inhibiting ATP consuming pathways such as fatty acid synthesis, protein synthesis, and potentially active transport¹⁸⁷ (see Fig. 5.15). AMPK also promotes cellular autophagy, an energy conserving survival mechanism in low energy states by inhibiting mTOR.^{188,189}

There is abundant AMPK expression in the kidney, but the understanding of its impact on energy metabolism and transport in the kidney is just emerging. The role of AMPK

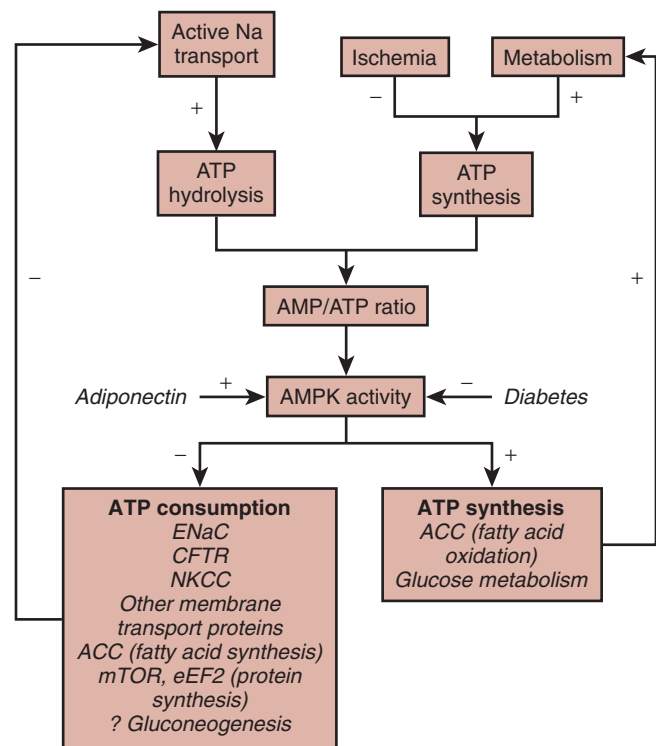


Fig. 5.15 Proposed role of adenosine monophosphate-activated protein kinase (AMPK) in the kidney in coupling catabolic pathways requiring ATP hydrolysis (primarily sodium transport) with metabolic pathways leading to ATP synthesis (primarily fatty acid and glucose oxidation). +, Activating pathway; –, inhibitory pathway. ACC, acetyl-CoA carboxylase. (From Hallows KR, et al. Role of the energy sensor AMP-activated kinase in renal physiology and disease. *Am J Physiol Renal Physiol.* 2010;298:F1067–F1077.)

in ion transport in the kidney has been recently reviewed.^{190,191} Overall, AMPK becomes activated when ATP is limiting, that is, when the AMP/ATP ratio increases, and once activated it decreases ATP consumption and increases ATP synthesis. In the kidney, sodium transport is the major energy-consuming process and there is increasing evidence regarding the role of AMPK in sodium transport in the kidney and other epithelial cells. AMPK has been shown to inhibit the activity of various transport proteins in the lung, gut, and kidney, including epithelial Na channel (ENaC) in the kidney collecting duct, Na-K-2Cl cotransporter (NKCC2) in the thick ascending limb, and Na⁺-K⁺-ATPase in the alveolar epithelial cells, particularly during hypoxia (as reviewed^{191,192}). Hallows and colleagues have determined that AMPK activation depresses transport mediated by the cystic fibrosis transmembrane conductance regulator, the epithelial sodium channel, the vacuolar H⁺-ATPase, and NKCC (see Fig. 5.15). A key question is whether renal AMPK activation suppresses Na⁺-K⁺-ATPase activity along the nephron. AMPK activation has been reported to inhibit lung cell Na⁺-K⁺-ATPase transport activity mediated by endocytosis,¹⁹³ but AMPK activation had no apparent effect on skeletal muscle Na⁺-K⁺-ATPase activity or distribution,¹⁹⁴ which leaves the question open for the kidney, where one report demonstrates that inhibition of AMPK induces the endocytosis of Na⁺-K⁺-ATPase in Madin-Darby canine kidney cells.¹⁹⁵

AMPK expression in the kidney is seen mainly in the cortical thick ascending limb and macula densa cells, and in some distal convoluted tubules and collecting ducts.¹⁹⁶ Recently, AMPK expression in the proximal tubule has also been reported.¹⁹⁷ Differing results in pAMPK expression in response to high salt intake has been observed with increased pAMPK in one study¹⁹⁶ and reduced expression in another.¹⁹⁸ In vivo, pharmacologic activation of AMPK in rats fed a high salt diet was shown to enhance TGF response and reduce sodium reabsorption in the proximal and distal tubule but had no effects on these parameters in rats fed a normal salt diet.¹⁹⁸ Whether these effects are driven by a change in cellular metabolism that results in an increase in AMP/ATP levels is not clear. Consumption of a high salt diet does decrease the fraction of the filtered load of Na⁺ that is absorbed. Given the effect of AMPK activation in inhibiting transporters, the findings suggest that AMPK participates in salt and water homeostasis. The AMPK pathway may provide another important layer of regulation between ATP production by mitochondria and ATP consumption by transporters. In the Whittam model illustrated in Fig. 5.3, increased active transport provokes a decrease in the ATP/ADP ratio, which drives increased ATP production by the mitochondria. When ATP production by the mitochondria becomes limiting, however, AMPK is likely to be activated, which would drive a reduction in ATP consumption by active membrane transporters. Thus, AMPK may regulate the coupling of ion transport and energy metabolism in the kidney.

Regulation of glucose and lipid metabolism by AMPK in pathophysiologic conditions such as diabetic nephropathy and CKD has been elucidated in recent publications. In animal models of CKD, diabetes, and obesity, reduced AMPK activity has been described.^{199–202} Endogenous AMPK activation by targeting adiponectin (adipose tissue-derived cytokine) as well as pharmacologic AMPK activation (metformin, AICAR) has been shown to improve glucose and lipid

homeostasis in diabetes and obesity-associated kidney disease. In a recent paper, significant changes in renal metabolism with the deletion of a major upstream AMPK activator, liver kinase B1 (LKB1), in kidney distal tubules were observed.²⁰³ Expression of AMPK and other key regulators of metabolism were significantly diminished and accompanied by tubular epithelial injury and interstitial fibrosis. In cultured epithelial cells, loss of LKB1 was associated with diminished fatty acid oxidation and glycolysis, leading to energy depletion and apoptotic cell death. In human kidney samples, lower levels of phosphorylated LKB1 and AMPK α 2 subunit were seen in CKD patients. Thus, the important role of AMPK in regulating kidney metabolism is being increasingly recognized.

Acute renal ischemia provokes a rapid and powerful activation of AMPK, but its functional role in the response to ischemia remains unclear. There is conflicting evidence in the literature regarding the effects of AMPK activation in renal ischemia–reperfusion injury.^{204–207} Similarly, while several studies show a beneficial effect of AMPK activation in myocardial ischemia–reperfusion, there are some conflicting studies demonstrating deleterious effects of AMPK activation in ischemic injury in the heart and brain (as reviewed¹⁸⁵). Hence, the effects of AMPK activation are likely to be time, tissue, and cell-dependent. Whether AMPK abundance or its phosphorylation is higher in the hypoxia-prone medulla than in the cortex has not yet been investigated, nor have studies been conducted examining the effect of AMPK activation on renal gluconeogenesis or glycolysis.^{193–195}

TUBULAR METABOLISM

Until recently there were limited studies exploring cellular metabolism in pathophysiologic conditions. Several recent publications have highlighted metabolic alterations in various forms of CKD. Moreover, alterations in glucose and fatty acid metabolism as early events in the course of diabetic and other forms of CKD place metabolic reprogramming central to the pathophysiology of CKD.

In mouse models of autosomal dominant polycystic kidney disease (ADPKD) and in kidney tissue from patients with ADPKD, defective glucose metabolism—specifically, increased aerobic glycolysis—was observed.²⁰⁸ Moreover, treatment with 2-deoxyglucose, an inhibitor of glycolysis, lowered kidney weight, volume, cystic index, and proliferation rates in mouse models of ADPKD.²⁰⁹ This phenotypic feature of enhanced aerobic glycolysis, also known as Warburg effect, is typically seen in proliferating cancer cells. In these mice models of PKD1 gene inactivation, there was inhibition of liver kinase B1 (LKB1)-AMPK axis along with activation of mTOR complex I pathway, leading to increased glycolysis.²⁰⁸

Besides altered glucose metabolism, changes in fatty acid metabolism in the kidney have been described in CKD. In a recent paper, Kang et al. described reduced levels of enzymes and regulators involved in fatty acid oxidation in mouse models of fibrosis.²¹⁰ In transcriptional analyses of microdissected human kidney samples from patients with diabetic or hypertensive CKD changes in fatty acid metabolism, β -oxidation, amino acid catabolism, and carbohydrate metabolism were observed. Genes related to fatty acid metabolism and their key transcriptional regulator complex, PPARA–PPARGC1A, were markedly lower in CKD samples compared with samples obtained from people with normal kidney function. This was

associated with higher lipid accumulation in renal tubular epithelial cells. Key regulators of glucose utilization were also lower in CKD samples. Using elegant experiments with genetic and pharmacologic approaches, the reduction in fatty acid oxidation was shown to be directly implicated in the pathogenesis of renal fibrosis in CKD. Although lipid accumulation in tubular cells in CKD patients has been reported before, these findings showed convincingly that it is the reduction in fatty acid oxidation rather than just the accumulation of lipid in tubular cells that plays a role in the development of renal fibrosis.

Metabolic reprogramming has also been recently reported in diabetic CKD.²¹¹ As discussed earlier, proximal tubules are largely dependent on mitochondrial oxidative metabolism for energy with limited glycolytic capacity. Under physiologic conditions, proximal tubules mainly use fatty acids, lactate, and glutamine as substrates for energy generation. In diabetes, there are increases in plasma concentration of glucose and fatty acids. These changes are also found intracellularly in various tissues including the kidneys, particularly proximal tubules. In an elegant study, Sas et al. provided a comprehensive assessment of substrate metabolism in diabetes using a systems-based approach that included transcriptomics, metabolomics, and metabolic flux analyses to determine alterations in glucose and fatty acid metabolism.²¹¹ In the kidney cortex in a type 2 diabetic mouse model (db/db mice), transcriptomic and metabolomic profiling demonstrated an increase in glycolysis, fatty acid β -oxidation, and tricarboxylic acid (TCA) cycle flux. Increased renal metabolism was associated with increased protein acetylation, which is a nutrient sensing posttranslational modification. There was also evidence for mitochondrial dysfunction as discussed later. Remarkably, transcriptomic analysis of kidney biopsy samples in patients with type 2 diabetes identified significant enrichment of many pathways involved in fatty acid, glucose, and amino acid metabolism and network analysis showed results similar to those obtained in db/db mice. In another cohort of type 1 diabetic patients enrolled in the Finnish Diabetic Nephropathy (FinnDiane) Study, increased glycolytic (hexose-6-phosphates, 2,3-phosphoglycerate) and TCA cycle metabolites (succinate, fumarate, malate) were detected in the urine compared with healthy control subjects. Urine samples from a subset of patients from the Family Investigation of Nephropathy and Diabetes (FIND) study, showed increased glycolytic intermediates at baseline in diabetic subjects compared with control subjects. TCA cycle intermediates in the urine were also elevated in diabetic subjects and predicted diabetic kidney disease progression, demonstrating their potential as prognostic biomarkers of diabetic kidney disease progression. These and previously discussed studies highlight the adaptive and maladaptive metabolic reprogramming and its role in the pathophysiology of diabetic and nondiabetic CKD.

MITOCHONDRIAL DISORDERS

Given the central role of mitochondria in producing ATP via oxidative phosphorylation, it is not surprising that genetic mutations affecting mitochondrial function have renal manifestations. Moreover, several lines of investigation support the significant role of early mitochondrial dysfunction in the pathophysiology of acute and chronic kidney diseases. The

mitochondrial genome is distinct from the nuclear genome, and it encodes 13 of the 88 protein subunits of ETC complexes I through V as well as 22 mitochondrial-specific transfer RNAs (tRNAs) and two RNA components of the translational apparatus. The nuclear genome encodes the remaining respiratory chain subunits as well as most of the mitochondrial DNA replication and expression components.

Disorders affecting mitochondrial oxidative phosphorylation can arise from mutations in either mitochondrial genes or nuclear genes encoding respiratory chain components or ancillary factors involved in maintenance of ETC function or the overall number of mitochondria.²¹² The incidence of genetic mitochondrial disorders is estimated at about 1 in 5000 births, with the most common affecting the sequence of a mitochondrial tRNA for leucine. Such mutations affect mitochondrial function in all tissues. Symptoms are evident before 2 months of age, and the number of organ systems affected increases with age.

Impairment of mitochondrial oxidative phosphorylation results in increased levels of reducing equivalents (NADH, FADH), which, in the mitochondria, transform acetoacetate to 3-hydroxybutyrate and, in the cytosol, transform pyruvate to lactate. Thus, elevated levels of lactic acid, ketone bodies, and impaired redox status are suggestive of a mitochondrial defect disorder.²¹³ If the genetic cause of the impairment can be pinpointed, then an appropriate therapy, if available, can be implemented to treat these life-threatening disorders; for example, coenzyme Q₁₀ enzyme defects can be treated with coenzyme Q₁₀ supplementation.²¹²

Myopathies and cardiomyopathies are the most common manifestations of mitochondrial disease, and central nervous system symptoms, including encephalopathies, are very common. Renal system impairment can be present but is not seen without other system deficiencies and is usually reported in children. Although glomerular disease and tubulointerstitial nephropathy have both been reported, the most frequently observed is impairment of proximal tubule reabsorption, known as de Toni–Debré–Fanconi syndrome, in which there are urinary losses of bicarbonate, amino acids, glucose, phosphate, uric acid, potassium, and water. All of these symptoms can be explained by a lack of ATP to fuel Na⁺-K⁺-ATPase sufficiently to drive transepithelial transport. The symptoms can range from mild to more severe and present in the neonatal period in most patients. Biopsy specimens show tubular dilations, casts, dedifferentiation, and cellular vacuolization. At the cellular level there are enlarged mitochondria. Supplements of sodium bicarbonate, potassium, vitamin D, phosphorus, and water are called for if these symptoms are evident.^{212,213}

Significant changes in mitochondrial function in diabetic kidneys have been described and recently reviewed in detail.^{214,215} Early changes in mitochondrial energetics, increased mitochondrial fragmentation, and reduced levels of PGC1 α have been described in diabetic kidneys. In diabetic nephropathy, increased metabolism has been shown to be associated with mitochondrial dysfunction.²¹¹ Assessment of oxidative phosphorylation in mitochondria isolated from renal cortices of 24-week-old control and diabetic mice showed increased proton leak and diminished mitochondrial ATP production. Additionally, total mitochondrial capacity was decreased in diabetic mitochondria, and the expression of mitochondrial uncoupling protein 2 was increased fourfold.

In renal cortexes from diabetic mice, expression of proteins in complexes I, II, III, and complex IV subunit cytochrome *c* oxidase subunit 4 (COX4) were significantly decreased at 24 weeks. The results of these studies suggested that dysfunction of ETC proteins in mitochondria from diabetic kidneys lead to less efficient ATP production and compensatory increases in glucose and fatty acid metabolic flux. This may be the primary metabolic abnormality in the diabetic kidney cortex as it precedes other markers of renal injury. Another study compared urine metabolites in patients with diabetes, but with or without CKD and in healthy controls.²¹⁶ Bioinformatic analyses revealed that 12 of the 13 differentially expressed metabolites were linked to mitochondrial metabolism. Kidney sections in patients with diabetes and CKD showed lower expression of mitochondrial proteins and lower gene expression of PGC1 α , which is a major regulator of mitochondrial biogenesis. Urine exosomes in these patients also showed less mitochondrial DNA. Evidence of mitochondrial dysfunction in nondiabetic CKD is also emerging. Early changes in mitochondrial function and structure in animal models of CKD has been reported.²¹⁷ In patients with CKD, mRNA levels of several mitochondrial enzymes and transcription factors were found to be lower, although no differences in mitochondrial copy number was seen.²¹⁰ In another clinical study, mitochondrial DNA copy number was evaluated in peripheral blood in a population cohort.²¹⁸ Participants with higher mitochondrial DNA copy numbers had a lower rate of prevalent diabetes and lower risk of incident CKD, even after adjusting for various risk factors for CKD.

The role of mitochondrial dysfunction in AKI has received significant attention. Several publications have elucidated the underlying mechanisms of mitochondrial dysfunction in various etiologies of AKI (reviewed in details²¹⁵). In ischemic and nephrotoxic AKI, decreased mitochondrial mass, disruption of cristae, and significant mitochondrial swelling has been observed.²¹⁹ In ischemic AKI, increase in mitochondrial NADH and dissipation of mitochondrial membrane potential in proximal tubules has been demonstrated.²²⁰ In ischemic and myoglobinuric AKI, reduction in various ETC proteins in the proximal tubule was shown.²²¹ Other studies have uncovered the crucial role of mitochondrial biogenesis using stimulators of PGC1 α , master regulator of mitochondrial biogenesis in recovery from ischemic AKI²²² and sepsis-associated AKI.¹⁵⁸ The role of mitochondrial dynamics has also been elucidated in some forms of AKI. In an elegant study, Brooks et al. described the disruption of mitochondrial dynamics and its pathogenic role in ischemic and nephrotoxic AKI.²¹⁹ Mitochondrial fragmentation was shown to precede tubular cell injury and death and pharmacologic inhibition of Drp1 prevented fragmentation and ameliorated AKI. Further highlighting the role of mitochondrial dysfunction, several pharmacologic targets to improve mitochondrial function in AKI and CKD have been investigated and found to be effective.²²³

SUMMARY

Most of the energy consumed by the kidney is traceable to the energy requirements for sodium reabsorption. Although all sodium reabsorption is linked to Na⁺-K⁺-ATPase, efficiency is achieved by leveraging Na⁺-K⁺-ATPase into transepithelial chloride or voltage gradients that allow some sodium to be reabsorbed without passing through the Na⁺-K⁺-ATPase itself. ATP production in the proximal tubule is solely by aerobic metabolism, whereas the medullary segments have additional capacity to produce energy by glycolysis. Transport activity regulates metabolism, metabolism may be rate limiting for transport, and the efficiency of transport can be made to vary at multiple levels from back-leak permeability to the efficiency of mitochondrial respiration. With regard to metabolic autoregulation, the kidney faces a particular challenge because the usual mechanism for delivering more oxygen to the kidney also increases the demand for that oxygen. Several intermediaries have been identified as parts of the complex network of interactions between transport and metabolism that allow the kidney to meet this challenge while balancing the risk of hypoxia against the risk of oxygen toxicity. A partial list of these includes adenosine, nitric oxide, prostaglandins, angiotensin II, dopamine, succinate, uncoupling proteins, HIF, and AMPK. A multiscale systems model that incorporates these elements along with renal anatomy to recapitulate renal metabolism is expected in the future.



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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